University of the Philippines - Baguio

Archie Junio's Reference Notebook

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Contents

1. Data Structures 1
1.1. Segment Tree 1
1.2. Union Find 1
1.3. 2D Sparse Table 1
2. Dynamic Programming 1
2.1. Lichao Tree 1
2.2. Divide And Conquer 2
3. Geometry 2
3.1. Angles And Rotations 2
3.2. Distances 2
3.3. Intersections 2
3.3.1. Circle Line Intersection 2
3.3.2. Circle Circle Intersection 2
3.3.3. Line Intersect 2
3.4. Closest Pair 2
3.5. Convex Hull 2
3.5.1. Convex Hull 2
3.5.2. Convex Hull 3D 2
3.6. Areas 3
3.6.1. Triangle Area 3
3.6.2. Polygon Area 3
3.7. Dot Cross 3
3.8. Line Sweep Closest Pair 3
3.9. Point Projection 3
3.10. Great Circle Distance 3
3.11. Point In Polygon 3
3.12. Cut Polygon By Line 4
3.13. Triangle Centers 4
3.14. Convex Polygon Intersection 4
3.15. Pick Theorem Lattice 5
3.16. Minimum Enclosing Circle 5
3.17. Shamos Algorithm 5
3.18. Kd Tree 5
4. Graphs 6
4.1. Max Flow 6
4.1.1. Dinic 6
4.1.2. Edmonds Karp 6
4.1.3. Bipartite Matching 6
4.2. Shortest Path 6
4.2.1. Dijkstra 6
4.2.2. Bellman Ford 7
4.3. Minimum Spanning Tree 7
4.3.1. Kruskal Min Spanning Tree 7
4.3.2. Prim Min Spanning Tree 7
4.4. Least Common Ancestor 7
4.4.1. Binary Lifting Lca 7
4.4.2. Euler Lca 7
4.4.3. Tarjan Lca 8
4.5. Minimum Mean Weight Cycle 8
4.6. Strongly Connected Components 8
4.7. Heavy Light Decomposition 8
4.8. Euler Path/Cycle 9
4.8.1. Directed Euler 9
4.8.2. Undirected Euler 9
4.9. Tree Isomorphism 9
4.10. Centroid Decomposition 10
5. Math I Algebra 10
5.0.1. Generating Function Manager 10
5.0.2. Fast Fourier Transform 10
5.0.3. Fft Polynomial Multiplication 11
5.0.4. Number Theoretic Transform 11
5.0.5. Polynomial Long Division 12
5.0.6. Matrix Multiplication 12
5.0.7. Matrix Power 12
5.0.8. Gauss Jordan Matrix Determinant 12
6. Math II Combinatorics 13
6.0.1. Lucas Theorem 13
6.0.2. Catalan Numbers 13
6.0.3. Kth Permutation 13
6.0.4. Factoradics 13
7. Math III Number Theory 13
7.0.1. Gcd Subset Counter 13
7.0.2. Linear Diophantine 13
7.0.3. Chinese Remainder Theorem 13
7.0.4. Euler Totient 13
7.0.5. Extended Euclidean Algorithm 13
7.0.6. Modular Exponentiation 14
7.0.7. Mod Inverse 14
7.0.8. Modulo Solver 14
7.0.9. Primitive Root 14
7.0.10. Josephus 14
8. Math IV Numerical Methods 14
8.0.1. Fast Square Testing 14
8.0.2. Simpson Integration 14
9. Strings 14
9.0.1. Longest Common Prefix 14
9.0.2. Hasher 14
9.0.3. Z Algorithm 15
9.0.4. Aho Corasick Trie 15
9.0.5. Knuth Morris Pratt 15
9.0.6. Trie 15
9.0.7. Suffix Array 15
9.0.8. Booth Min String Rotation 16
10. Other Algorithms 16
10.0.1. Longest Increasing Subsequence 16
10.0.2. Cycle Find 16
10.0.3. 2Sat 16
10.0.4. Dpll 17
10.0.5. Stable Marriage 17
10.0.6. Dates 17
10.0.7. Simulated Annealing 17
10.0.8. Fast Input Reading 18
10.0.9. Bit Hacks 18

1. Data Structures

1.1. Segment Tree.

```
class SegmentTree:
    def __init__(self, data):
        self.n = len(data)
        self.tree = [0] * (2 * self.n)
        for i in range(self.n):
            self.tree[self.n + i] = data[i]
        for i in range(self.n - 1, 0, -1):
            self.tree[i] = self.tree[i << 1] + self.tree[i << 1 | 1]

    def update(self, p, value):
        p += self.n
        self.tree[p] = value
        while p > 1:
            self.tree[p >> 1] = self.tree[p] + self.tree[p ^ 1]
            p >>= 1

    def query(self, l, r):
        res = 0
        l += self.n
        r += self.n
        while l < r:
            if l & 1:
                res += self.tree[l]
                l += 1
            if r & 1:
                r -= 1
                res += self.tree[r]
            l >>= 1
            r >>= 1
        return res
```

1.2. Union Find.

```
class DSU:
    def __init__(self, n):
        self.parent = list(range(n))
        self.size = [1] * n

    def find(self, i):
        if self.parent[i] == i:
```

```
---------- return i ----------------------------------------------------------
------- self.parent[i] = self.find(self.parent[i]) ---------------------------
------- return self.parent[i] ------------------------------------------------
 ------------------------------------------------------------------------------
--- def union(self, i, j): -------------------------------------------------
------- root_i = self.find(i) ------------------------------------------------
------- root_j = self.find(j) ------------------------------------------------
------- ---------------------------------------------------------------------
------- if root_i != root_j: -------------------------------------------------
---------- if self.size[root_i] < self.size[root_j]: -------------------------
-------------- root_i, root_j = root_j, root_i ---------------------------------
---------- self.parent[root_j] = root_i ---------------------------------------
---------- self.size[root_i] += self.size[root_j] -----------------------------
---------- return True --------------------------------------------------------
------- return False ---------------------------------------------------------
 ------------------------------------------------------------------------------
--- def get_size(self, i): -------------------------------------------------
------- return self.size[self.find(i)] ---------------------------------------
```

1.3. 2D Sparse Table.

```
import math ------------------------------------------------------------------
 ------------------------------------------------------------------------------
class SparseTable2D: ---------------------------------------------------------
--- def __init__(self, A): -------------------------------------------------
------- self.A = A -----------------------------------------------------------
------- self.n = len(A) ------------------------------------------------------
------- self.m = len(A[0]) ---------------------------------------------------
 ------------------------------------------------------------------------------
------- self.LOGN = int(math.log2(self.n)) + 1 -------------------------------
------- self.LOGM = int(math.log2(self.m)) + 1 -------------------------------
 ------------------------------------------------------------------------------
------- self.lg = [0] * (max(self.n, self.m) + 1) ----------------------------
------- for i in range(2, len(self.lg)): -------------------------------------
---------- self.lg[i] = self.lg[i // 2] + 1 ----------------------------------
 ------------------------------------------------------------------------------
------- self.st = [[[[0] * self.m for _ in range(self.n)] ------------------
------------------ for _ in range(self.LOGM)] --------------------------------
----------------- for _ in range(self.LOGN)] --------------------------------
 ------------------------------------------------------------------------------
------- self.build() ---------------------------------------------------------
 ------------------------------------------------------------------------------
--- def build(self): -------------------------------------------------------
------- for i in range(self.n): ----------------------------------------------
---------- for j in range(self.m): -------------------------------------------
--------------- self.st[0][0][i][j] = self.A[i][j] -------------------------
 ------------------------------------------------------------------------------
------- for bj in range(1, self.LOGM): ---------------------------------------
---------- for i in range(self.n): -------------------------------------------
--------------- for j in range(self.m) -------------------------------------
------------------ self.st[0][bj][i][j] = max( ------------------------------
---------------------- self.st[0][bj - 1][i][j], -------------------------
---------------------- self.st[0][bj - 1][i][j + (1 << (bj - 1))] ---
------------------ ) -------------------------------------------------------
 ------------------------------------------------------------------------------
------- for bi in range(1, self.LOGN): ---------------------------------------
---------- for i in range(self.n - (1 << bi) + 1): ---------------------------
--------------- for j in range(self.m): ------------------------------------
------------------ self.st[bi][0][i][j] = max( ------------------------------
---------------------- self.st[bi - 1][0][i][j], -------------------------
---------------------- self.st[bi - 1][0][i + (1 << (bi - 1))][j] ---
------------------ ) -------------------------------------------------------
 ------------------------------------------------------------------------------
------- for bi in range(1, self.LOGN): ---------------------------------------
---------- for bj in range(1, self.LOGM): ------------------------------------
--------------- for i in range(self.n - (1 << bi) + 1): --------------------
------------------ for j in range(self.m - (1 << bj) + 1): -----------------
---------------------- self.st[bi][bj][i][j] = max( ----------------------
-------------------------- self.st[bi - 1][bj - 1][i][j], ------------------
-------------------------- self.st[bi - 1][bj - 1][i + (1 << (bi -
↪  1))][j],
-------------------------- self.st[bi - 1][bj - 1][j + (1 << (bj -
↪  1))],
-------------------------- self.st[bi - 1][bj - 1][i + (1 << (bi -
↪  1))][j + (1 << (bj - 1))]
---------------------- ) -----------------------------------------------------
 ------------------------------------------------------------------------------
--- def query(self, x1, y1, x2, y2): ---------------------------------------
------- kx = self.lg[x2 - x1 + 1] --------------------------------------------
------- ky = self.lg[y2 - y1 + 1] --------------------------------------------
 ------------------------------------------------------------------------------
------- return max( ----------------------------------------------------------
---------- self.st[kx][ky][x1][y1], ------------------------------------------
---------- self.st[kx][ky][x2 - (1 << kx) + 1][y1], --------------------------
---------- self.st[kx][ky][x1][y2 - (1 << ky) + 1], --------------------------
---------- self.st[kx][ky][x2 - (1 << kx) + 1][y2 - (1 << ky) + 1] --
------- ) --------------------------------------------------------------------
# Input Format: st = SparseTable2D(grid) ------------------------------------
#               print(st.query(1, 1, 2, 2)) ------------------------------------
```

2. Dynamic Programming

2.1. Lichao Tree.

```
class Line: ------------------------------------------------------------------
--- __slots__ = ['m', 'c'] -------------------------------------------------
--- def __init__(self, m=0, c=float('-inf')): ------------------------------
------- self.m = m -----------------------------------------------------------
------- self.c = c -----------------------------------------------------------
------- ---------------------------------------------------------------------
--- def eval(self, x): -----------------------------------------------------
------- return self.m * x + self.c -------------------------------------------
 ------------------------------------------------------------------------------
class Node: ------------------------------------------------------------------
--- __slots__ = ['line', 'left', 'right'] ----------------------------------
--- def __init__(self): ----------------------------------------------------
------- self.line = Line() ---------------------------------------------------
------- self.left = None -----------------------------------------------------
------- self.right = None ----------------------------------------------------
 ------------------------------------------------------------------------------
class LiChaoTree: ------------------------------------------------------------
--- def __init__(self, min_x=-10**12, max_x=10**12): -----------------------
------- self.root = Node() ---------------------------------------------------
------- self.min_x = min_x ---------------------------------------------------
------- self.max_x = max_x ---------------------------------------------------
 ------------------------------------------------------------------------------
--- def add_line(self, m, c): ----------------------------------------------
------- self._insert(self.root, self.min_x, self.max_x, Line(m, c)) --
 ------------------------------------------------------------------------------
--- def _insert(self, node, l, r, new_line): -------------------------------
------- mid = (l + r) // 2 ---------------------------------------------------
------- ---------------------------------------------------------------------
------- left_better = new_line.eval(l) > node.line.eval(l) -------------------
------- mid_better = new_line.eval(mid) > node.line.eval(mid) ----------------
------- ---------------------------------------------------------------------
------- if mid_better: -------------------------------------------------------
---------- node.line, new_line = new_line, node.line -------------------------
------- if r - l <= 1: -------------------------------------------------------
---------- return ------------------------------------------------------------
---------- ------------------------------------------------------------------
------- if left_better != mid_better: ----------------------------------------
---------- if not node.left: -------------------------------------------------
--------------- node.left = Node() -----------------------------------------
---------- self._insert(node.left, l, mid, new_line) -------------------------
------- else: ----------------------------------------------------------------
---------- if not node.right: ------------------------------------------------
--------------- node.right = Node() ----------------------------------------
---------- self._insert(node.right, mid, r, new_line) ------------------------
 ------------------------------------------------------------------------------
--- def query(self, x): ----------------------------------------------------
------- return self._query(self.root, self.min_x, self.max_x, x) -----
 ------------------------------------------------------------------------------
--- def _query(self, node, l, r, x): ---------------------------------------
------- if not node: ---------------------------------------------------------
---------- return float('-inf') ----------------------------------------------
------- if r - l <= 1: -------------------------------------------------------
---------- return node.line.eval(x) ------------------------------------------
---------- ------------------------------------------------------------------
------- mid = (l + r) // 2 ---------------------------------------------------
------- res = node.line.eval(x) ----------------------------------------------
------- ---------------------------------------------------------------------
------- if x < mid: ----------------------------------------------------------
---------- res = max(res, self._query(node.left, l, mid, x)) ----------------
------- else: ----------------------------------------------------------------
---------- res = max(res, self._query(node.right, mid, r, x)) ----------------
---------- ------------------------------------------------------------------
------- return res -----------------------------------------------------------
```

2.2. Divide And Conquer.

```
INF = float('inf') -----------------------------------------------------------
 ------------------------------------------------------------------------------
def solve_dp(g, k_L, k_R, n_L, n_R, dp, cost_func): --------------------------
--- if n_L > n_R: ----------------------------------------------------------
------- return ---------------------------------------------------------------
 ------------------------------------------------------------------------------
--- n_M = (n_L + n_R) // 2 -------------------------------------------------
--- best_k = -1 ------------------------------------------------------------
--- dp[g][n_M] = INF -------------------------------------------------------
 ------------------------------------------------------------------------------
--- for k in range(k_L, min(n_M, k_R) + 1): --------------------------------
------- current_val = dp[g - 1][k] + cost_func(k + 1, n_M) -------------------
------- if current_val < dp[g][n_M]: -----------------------------------------
---------- dp[g][n_M] = current_val ------------------------------------------
---------- best_k = k --------------------------------------------------------
 ------------------------------------------------------------------------------
--- solve_dp(g, k_L, best_k, n_L, n_M - 1, dp, cost_func) ------------------
--- solve_dp(g, best_k, k_R, n_M + 1, n_R, dp, cost_func) ------------------
```

3. Geometry

3.1. Angles And Rotations.

```
def angle(a, b, c): # angle formed by abc in radians: PI < x <= PI ---
--- return abs((arg(a-b) - arg(c-b)) % (2 * math.pi)) ----------------------
def rotate(p, a, d): # rotate point a about pivot p CCW at d radians -
--- return p + (a - p) * complex(math.cos(d), math.sin(d)) -----------------
```

3.2. Distances.

```
 ------------------------------------------------------------------------------
# Distance: Point to 2D Line (ax + by + c = 0) ---------------------------
def dist_pt_line_eq(p, a, b, c): ---------------------------------------------
--- return abs(a*p.real + b*p.imag + c) / math.hypot(a, b) -----------------
 ------------------------------------------------------------------------------
# Distance: Point to 2D Line through points a and b ----------------------
def dist_pt_line(p, a, b): ---------------------------------------------------
--- cp = (p - a).conjugate() * (b - a) -------------------------------------
--- return abs(cp.imag) / abs(b - a) ---------------------------------------
 ------------------------------------------------------------------------------
# Distance: Point to 3D Plane (ax + by + cz + d = 0) ---------------------
def dist_pt_plane(p, a, b, c, d): --------------------------------------------
--- # math.hypot supports 3+ arguments in Python 3.8+ ------------------
--- return abs(a*p.x + b*p.y + c*p.z + d) / math.hypot(a,b,c) --------
 ------------------------------------------------------------------------------
# Distance: Shortest distance between 3D lines AB and CD -------------
def dist_line_3d(A, B, C, D): ------------------------------------------------
--- u, v, w = B - A, D - C, A - C ------------------------------------------
--- a, b = u.dot(u), u.dot(v) ----------------------------------------------
--- c, d, e = v.dot(v), u.dot(w), v.dot(w) ---------------------------------
--- det = a * c - b * b ----------------------------------------------------
--- if det < 1e-9: # Parallel lines ----------------------------------------
------- s, t = 0.0, d / b if b > c else e / c ----------------------------
--- else: # Skew lines -----------------------------------------------------
------- s = (b * e - c * d) / det --------------------------------------
------- t = (a * e - b * d) / det --------------------------------------
```

```
--- p1 = P3(A.x + u.x*s, A.y + u.y*s, A.z + u.z*s) ---------------------
--- p2 = P3(C.x + v.x*t, C.y + v.y*t, C.z + v.z*t) ---------------------
--- vec = p1 - p2 ------------------------------------------------------
--- return math.sqrt(vec.dot(vec)) -------------------------------------
```

3.3. Intersections.

3.3.1. *Circle Line Intersection.*

```
EPS = 1e-9 ---------------------------------------------------------------
def projLine(p, a, b): # Project point p onto line segment ab --------
--- v = b - a ----------------------------------------------------------
--- scalar = ((p - a) * v.conjugate()).real / abs(v)**2 ---------------
--- return a + v * scalar ----------------------------------------------
 ------------------------------------------------------------------------
def rotate(c, p, t): # Rotate point p around center c by t radians ---
--- return c + (p - c) * cmath.rect(1, t) ------------------------------
 ------------------------------------------------------------------------
def CL_inter(c, r, a, b): ------------------------------------------------
--- p = projLine(c, a, b) ----------------------------------------------
--- d = abs(c - p) -----------------------------------------------------
--- ans = [] -----------------------------------------------------------
--- ------------------------------------------------------------------
--- if d > r + EPS: pass # None ----------------------------------------
--- elif d > r - EPS: ans.append(p) # Tangent -------------------------
--- elif d < EPS: # Diameter -------------------------------------------
------- v = r * (b - a) / abs(b - a) -------------------------------
------- ans.append(c + v) ------------------------------------------
------- ans.append(c - v) ------------------------------------------
--- else: # Secant -----------------------------------------------------
------- t = math.acos(d / r) ---------------------------------------
------- p = c + (p - c) * r / d ------------------------------------
------- ans.append(rotate(c, p, t)) --------------------------------
------- ans.append(rotate(c, p, -t)) -------------------------------
--- return ans ---------------------------------------------------------
```

3.3.2. *Circle Circle Intersection.*

```
EPS = 1e-9 ---------------------------------------------------------------
def rotate(c, p, t): -----------------------------------------------------
--- # Rotate point p around center c by t radians ---------------------
--- return c + (p - c) * cmath.rect(1, t) ------------------------------
 ------------------------------------------------------------------------
def CC_inter(c1, r1, c2, r2): --------------------------------------------
--- d = abs(c1 - c2) ---------------------------------------------------
--- ans = [] -----------------------------------------------------------
--- if d < EPS: # Concentric circles -----------------------------------
------- if abs(r1 - r2) < EPS: -------------------------------------
----------- pass # Infinite intersections (same circle) --------------
--- elif r1 < EPS: # Circle 1 is a point -------------------------------
------- if abs(d - r2) < EPS: --------------------------------------
----------- ans.append(c1) -------------------------------------------
--- else: # General case using Law of Cosines ---------------------------
------- s = (r1*r1 + d*d - r2*r2) / (2*r1*d) -----------------------
------- s = max(-1.0, min(1.0, s)) # Prevent float domain errors -----
------- t = math.acos(s) -------------------------------------------
------- # Point on C1's radius pointing towards C2 -------------------
------- mid = c1 + (c2 - c1) * r1 / d ------------------------------
------- ans.append(rotate(c1, mid, t)) -----------------------------
------- if abs(math.sin(t)) >= EPS: --------------------------------
----------- # FIXED: Rotate around c1, not c2! ----------------------
----------- ans.append(rotate(c1, mid, -t)) --------------------------
--- return ans ---------------------------------------------------------
```

3.3.3. *Line Intersect.*

```
def line_inter_fast(a, b, c, d): -----------------------------------------
--- A = cross(d - a, b - a) --------------------------------------------
--- B = cross(c - a, b - a) --------------------------------------------
--- return (B * d - A * c) / (B - A) if B - A else None ----------------
```

3.4. Closest Pair.

```
def closest_pair(pts): ---------------------------------------------------
--- if len(pts) <= 1: return float('inf') ------------------------------
--- px = sorted(pts) # Automatically sorts by x, then y ----------------
--- def solve(l, r): # Base case: Brute force for small partitions ---
------- if r - l <= 3: ---------------------------------------------
----------- d = float('inf') -----------------------------------------
----------- for i in range(l, r): ------------------------------------
--------------- for j in range(i + 1, r): ------------------------------
------------------- d = min(d, math.dist(px[i], px[j])) --------------
----------- return d -------------------------------------------------
------- mid = (l + r) // 2 -----------------------------------------
------- mid_x = px[mid][0] -----------------------------------------
------- # Recursively find the minimum in both halves ----------------
------- d = min(solve(l, mid), solve(mid, r)) ----------------------
------- # Gather points in the middle strip ---------------------------
------- strip = [p for p in px[l:r] if abs(p[0] - mid_x) < d] --------
------- strip.sort(key=lambda p: p[1]) # Sort strip by y -------------
------- # Check at most 7 geometric neighbors in the strip -----------
------- for i, p1 in enumerate(strip): -----------------------------
----------- for p2 in strip[i+1 : i+8]: ------------------------------
--------------- d = min(d, math.dist(p1, p2)) ----------------------
------- return d ---------------------------------------------------
--- return solve(0, len(px)) -------------------------------------------
```

3.5. Convex Hull.

3.5.1. *Convex Hull.*

```
def cross(o, a, b): # 2D cross product of OA and OB vectors ----------
--- return (a[0]-o[0])*(b[1]-o[1]) - (a[1]-o[1])*(b[0]-o[0]) ---------
 ------------------------------------------------------------------------
def convex_hull(pts): # Sorts by x, then by y automatically ----------
--- pts = sorted(set(pts))  --------------------------------------------
--- if len(pts) <= 1: --------------------------------------------------
------- return pts -------------------------------------------------
--- def build(iterable): -----------------------------------------------
------- h = [] -----------------------------------------------------
------- for p in iterable: -----------------------------------------
----------- while len(h) >= 2 and cross(h[-2], h[-1], p) <= 0: -------
--------------- h.pop() --------------------------------------------
----------- h.append(p) ----------------------------------------------
------- return h ---------------------------------------------------
--- lower = build(pts) -------------------------------------------------
--- upper = build(reversed(pts)) ---------------------------------------
--- return lower[:-1] + upper[:-1] -------------------------------------
```

3.5.2. *Convex Hull 3D.*

```
class P3: ----------------------------------------------------------------
--- __slots__ = ['x', 'y', 'z'] ----------------------------------------
--- def __init__(self, x=0, y=0, z=0): ---------------------------------
------- self.x, self.y, self.z = x, y, z ---------------------------
--- def __sub__(self, o): ----------------------------------------------
------- return P3(self.x-o.x, self.y-o.y, self.z-o.z) --------------
--- def cross(self, o): ------------------------------------------------
------- return P3(self.y*o.z - self.z*o.y, -------------------------
----------------- self.z*o.x - self.x*o.z, -------------------------
----------------- self.x*o.y - self.y*o.x) -------------------------
--- def dot(self, o): --------------------------------------------------
------- return self.x*o.x + self.y*o.y + self.z*o.z ----------------
--- def __eq__(self, o): -----------------------------------------------
------- return (self.x, self.y, self.z) == (o.x, o.y, o.z) ---------
--- def __lt__(self, o): -----------------------------------------------
------- return (self.x, self.y, self.z) < (o.x, o.y, o.z) ----------
 ------------------------------------------------------------------------
class Face: --------------------------------------------------------------
--- __slots__ = ['p', 'q'] ---------------------------------------------
--- def __init__(self, p, q): ------------------------------------------
------- self.p = p # Tuple of 3 point indices ----------------------
------- self.q = q # Normal vector (P3 object) ---------------------
 ------------------------------------------------------------------------
```

```
def hull_3D(pts): --------------------------------------------------------
--- n = len(pts) -------------------------------------------------------
--- faces = [] ---------------------------------------------------------
--- dead = [[True] * n for _ in range(n)] ------------------------------
--- def add_face(a, b, c): ---------------------------------------------
------- q = (pts[b] - pts[a]).cross(pts[c] - pts[a]) ---------------
------- faces.append(Face((a, b, c), q)) ---------------------------
------- dead[a][b] = dead[b][c] = dead[c][a] = False ---------------
--- # Initialize with a degenerate tetrahedron ---------------------
--- add_face(0, 1, 2) --------------------------------------------------
--- add_face(0, 2, 1) --------------------------------------------------
--- for i in range(3, n): ----------------------------------------------
------- f_inv = [] -------------------------------------------------
------- for f in faces: --------------------------------------------
----------- # If face is visible from the point ----------------------
----------- if (pts[i] - pts[f.p[0]]).dot(f.q) > 0: ------------------
--------------- for j in range(3): --------------------------------
------------------- dead[f.p[j]][f.p[(j+1)%3]] = True ---------------
----------- else: ----------------------------------------------------
--------------- f_inv.append(f) ------------------------------------
------- faces = [] -------------------------------------------------
------- for f in f_inv: --------------------------------------------
----------- for j in range(3): ---------------------------------------
--------------- a, b = f.p[j], f.p[(j+1)%3] ------------------------
--------------- # If reverse edge is dead but forward is alive -------
--------------- if dead[b][a]: ------------------------------------
------------------- add_face(b, a, i) -------------------------------
------- faces.extend(f_inv) ----------------------------------------
--- return faces -------------------------------------------------------
```

3.6. Areas.

3.6.1. *Triangle Area.*

```
def area(a, b, c): -------------------------------------------------------
--- s = (a + b + c) / 2 ------------------------------------------------
--- return (s*(s-a)*(s-b)*(s-c)) ** 0.5 --------------------------------
```

3.6.2. *Polygon Area.*

```
def area(p, n): ----------------------------------------------------------
--- a = 0 --------------------------------------------------------------
--- for i in range(n): -------------------------------------------------
------- j = (i - 1) % n --------------------------------------------
------- a += cross(p[i], p[j]) -------------------------------------
--- return abs(a) / 2 --------------------------------------------------
```

3.7. Dot Cross.

```
def dot(a, b): -----------------------------------------------------------
--- return a.x * b.x + a.y * b.y # + a.z * b.z -------------------------
def cross(a, b): ---------------------------------------------------------
--- return a.x * b.y - a.y * b.x ---------------------------------------
def cross(a, b, c): ------------------------------------------------------
--- return cross(a, b) + cross(b, c) + cross(c, a) ---------------------
def cross3D(a, b): -------------------------------------------------------
--- return point3(a.x*b.y - a.y*b.x, a.y*b.z - a.z*b.y, a.z*b.x -
↪   a.x*b.z)
```

3.8. Line Sweep Closest Pair.

```
import math

def dist_sq(p1, p2):
    return (p1[0] - p2[0])**2 + (p1[1] - p2[1])**2


def closest_pair(points):
    points.sort()

    def solve(pts):
        n = len(pts)

        if n <= 3:
            return min(dist_sq(pts[i], pts[j])
                       for i in range(n)
                       for j in range(i + 1, n)) if n > 1 else
↪   float('inf')

        mid = n // 2
        mid_x = pts[mid][0]

        left = pts[:mid]
        right = pts[mid:]

        d = min(solve(left), solve(right))

        strip = [p for p in pts if (p[0] - mid_x) ** 2 < d]
        strip.sort(key=lambda p: p[1])

        for i in range(len(strip)):
            for j in range(i + 1, min(i + 7, len(strip))):
                d = min(d, dist_sq(strip[i], strip[j]))

        return d

    return solve(points)
```

3.9. Point Projection.

```
EPS = 1e-9

def dot(a, b):
    return sum(x * y for x, y in zip(a, b))


def scale(v, k):
    return [k * x for x in v]


def add(a, b):
    return [x + y for x, y in zip(a, b)]


def sub(a, b):
    return [x - y for x, y in zip(a, b)]


def norm(v):
    return dot(v, v)

def proj(p, v):
    v_norm = norm(v)
    if v_norm < EPS:
        return [0] * len(p)
    return scale(v, dot(p, v) / v_norm)

def proj_line(p, a, b):
    ab = sub(b, a)
    ab_norm = norm(ab)
    if ab_norm < EPS:
        return a[:]

    t = dot(sub(p, a), ab) / ab_norm
    return add(a, scale(ab, t))

def proj_seg(p, a, b):
    ab = sub(b, a)
    ab_norm = norm(ab)
    if ab_norm < EPS:
        return a[:]

    t = dot(sub(p, a), ab) / ab_norm
    t = max(0.0, min(1.0, t))

    return add(a, scale(ab, t))

def proj_plane(p, a, b, c, d):
    n = [a, b, c]
    n_norm = norm(n)
    if n_norm < EPS:
        return p[:]

    dist = (dot(p, n) + d) / n_norm
    return sub(p, scale(n, dist))
```

3.10. Great Circle Distance.

```
import math

PI = math.pi

def clamp(x, lo, hi):
    return max(lo, min(hi, x))

# ========================================================
# version 1: lat/long (spherical law of cosines)
# ========================================================
def great_circle_dist_latlong(lat1, lon1, lat2, lon2, R=6371.0):
    lat1 *= PI / 180
    lon1 *= PI / 180
    lat2 *= PI / 180
    lon2 *= PI / 180

    val = (
        math.sin(lat1) * math.sin(lat2)
        + math.cos(lat1) * math.cos(lat2) * math.cos(abs(lon1 - lon2))
    )

    val = clamp(val, -1.0, 1.0)  # IMPORTANT FIX

    return R * math.acos(val)

def dot(a, b):
    return sum(x * y for x, y in zip(a, b))

def cross(a, b):
    return [
        a[1] * b[2] - a[2] * b[1],
        a[2] * b[0] - a[0] * b[2],
        a[0] * b[1] - a[1] * b[0],
    ]

def norm(v):
    return math.sqrt(dot(v, v))

# ========================================================
# version 2: using 3D vectors (angle between vectors)
# ========================================================
def great_circle_dist_vec(a, b, R=1.0):
    # assumes a and b are unit vectors on sphere
    cross_prod = cross(a, b)
    return R * math.atan2(norm(cross_prod), dot(a, b))

def to_vec(lat, lon):
    lat *= PI / 180
    lon *= PI / 180

    return [
        math.cos(lat) * math.cos(lon),
        math.cos(lat) * math.sin(lon),
        math.sin(lat),
    ]
```

3.11. Point In Polygon.

```
import math

EPS = 1e-9

def dist(a, b):
    return math.hypot(a[0] - b[0], a[1] - b[1])

def in_polygon(q, p):
    n = len(p)
    inside = False

    j = n - 1
    for i in range(n):
        xi, yi = p[i]
        xj, yj = p[j]
        xq, yq = q

        intersect = ((yi > yq) != (yj > yq)) and (
            xq < (xj - xi) * (yq - yi) / (yj - yi + EPS) + xi
        )

        if intersect:
            inside = not inside

        j = i

    return inside

def on_polygon(q, p):
    n = len(p)

    for i in range(n):
        j = (i + 1) % n

        a = p[i]
        b = p[j]

        if abs(dist(a, q) + dist(q, b) - dist(a, b)) < EPS:
            return True

    return False
```

3.12. Cut Polygon By Line.

```
EPS = 1e-9

def cross(a, b, c):
    return (b[0] - a[0]) * (c[1] - a[1]) - (b[1] - a[1]) * (c[0] -
↪   a[0])


def line_inter(p1, p2, a, b):
    x1, y1 = p1
    x2, y2 = p2
    x3, y3 = a
    x4, y4 = b

    den = (x1 - x2) * (y3 - y4) - (y1 - y2) * (x3 - x4)
    if abs(den) < EPS:
        return p1
```

```
--------------------------------------------------------------
--- px = (
------- (x1 * y2 - y1 * x2) * (x3 - x4)
------- - (x1 - x2) * (x3 * y4 - y3 * x4)
--- ) / den
--------------------------------------------------------------
--- py = (
------- (x1 * y2 - y1 * x2) * (y3 - y4)
------- - (y1 - y2) * (x3 * y4 - y3 * x4)
--- ) / den
--------------------------------------------------------------
--- return (px, py)
--------------------------------------------------------------
def cut(poly, a, b):
--- n = len(poly)
--- res = []
--------------------------------------------------------------
--- for i in range(n):
------- j = (i - 1 + n) % n
--------------------------------------------------------------
------- c1 = cross(a, b, poly[j])
------- c2 = cross(a, b, poly[i])
--------------------------------------------------------------
------- if c1 >= -EPS:
----------- res.append(poly[j])
--------------------------------------------------------------
------- if c1 * c2 < -EPS:
----------- inter = line_inter(poly[j], poly[i], a, b)
----------- res.append(inter)
--------------------------------------------------------------
--- return res
```

3.13. Triangle Centers.

```
import math
--------------------------------------------------------------
EPS = 1e-9
--------------------------------------------------------------
def add(a, b):
--- return (a[0] + b[0], a[1] + b[1])
def sub(a, b):
--- return (a[0] - b[0], a[1] - b[1])
def scale(a, k):
--- return (a[0] * k, a[1] * k)
def norm2(a, b):
--- return (a[0] - b[0]) ** 2 + (a[1] - b[1]) ** 2
def bary(A, B, C, a, b, c):
--- s = a + b + c
--- if abs(s) < EPS:
------- return (0.0, 0.0)
--------------------------------------------------------------
--- return (
------- (A[0] * a + B[0] * b + C[0] * c) / s,
------- (A[1] * a + B[1] * b + C[1] * c) / s,
--- )
def dist(a, b):
--- return math.sqrt(norm2(a, b))
def trilinear(A, B, C, a, b, c):
--- return bary(
------- A,
------- B,
------- C,
------- dist(B, C) * a,
------- dist(C, A) * b,
------- dist(A, B) * c,
--- )
def centroid(A, B, C):
--- return bary(A, B, C, 1, 1, 1)
def circumcenter(A, B, C):
--- a = dist(B, C)
--- b = dist(C, A)
--- c = dist(A, B)
--------------------------------------------------------------
--- return bary(
------- A,
------- B,
------- C,
------- a * (b + c - a),
------- b * (c + a - b),
------- c * (a + b - c),
--- )
def orthocenter(A, B, C):
--- def foot(P, Q, R):
------- q = sub(Q, R)
------- t = dot(sub(P, R), q) / (dot(q, q) + EPS)
------- return add(R, scale(q, t))
--- def dot(a, b):
------- return a[0] * b[0] + a[1] * b[1]
--------------------------------------------------------------
--- H1 = foot(A, B, C)
--- H2 = foot(B, A, C)
--------------------------------------------------------------
--- return bary(A, B, C, 1, 1, 1)
def incenter(A, B, C):
--- return bary(
------- A,
------- B,
------- C,
------- dist(B, C),
------- dist(C, A),
------- dist(A, B),
--- )
def inradius(a, b, c):
--- s = (a + b + c) / 2
--- area = math.sqrt(max(s * (s - a) * (s - b) * (s - c), 0))
--- return area / s if s > EPS else 0
def excenter(A, B, C):
--- a = dist(B, C)
--- b = dist(C, A)
--- c = dist(A, B)
--- return bary(A, B, C, -a, b, c)
def symmedian(A, B, C):
--- return bary(
------- A,
------- B,
------- C,
------- norm2(B, C),
------- norm2(C, A),
------- norm2(A, B),
--- )
```

3.14. Convex Polygon Intersection.

```
EPS = 1e-9
--------------------------------------------------------------
def sub(a, b):
--- return (a[0] - b[0], a[1] - b[1])
def add(a, b):
--- return (a[0] + b[0], a[1] + b[1])
def cross(a, b):
--- return a[0] * b[1] - a[1] * b[0]
def dot(a, b):
--- return a[0] * b[0] + a[1] * b[1]
def orient(a, b, c):
--- return cross(sub(b, a), sub(c, a))
def seg_inter(a, b, c, d):
--- def on_seg(p, a, b):
------- return (
----------- abs(orient(a, b, p)) < EPS
----------- and min(a[0], b[0]) - EPS <= p[0] <= max(a[0], b[0]) + EPS
----------- and min(a[1], b[1]) - EPS <= p[1] <= max(a[1], b[1]) + EPS
------- )
--- def line_inter(a, b, c, d):
------- A1 = b[1] - a[1]
------- B1 = a[0] - b[0]
------- C1 = A1 * a[0] + B1 * a[1]
------- A2 = d[1] - c[1]
------- B2 = c[0] - d[0]
------- C2 = A2 * c[0] + B2 * c[1]
------- det = A1 * B2 - A2 * B1
------- if abs(det) < EPS:
----------- return None
------- x = (B2 * C1 - B1 * C2) / det
------- y = (A1 * C2 - A2 * C1) / det
------- return (x, y)
--- p = line_inter(a, b, c, d)
--- if p is None:
------- return None
--- if (
------- on_seg(p, a, b)
------- and on_seg(p, c, d)
--- ):
------- return p
--- return None
def in_poly(p, poly):
--- inside = False
--- n = len(poly)
--- for i in range(n):
------- j = (i - 1) % n
------- a, b = poly[i], poly[j]
------- if ((a[1] > p[1]) != (b[1] > p[1])):
----------- x = (b[0] - a[0]) * (p[1] - a[1]) / (b[1] - a[1] + EPS) +
↪  a[0]
----------- if p[0] < x:
--------------- inside = not inside
--- return inside
def on_poly(p, poly):
--- n = len(poly)
--- for i in range(n):
------- j = (i - 1) % n
------- if abs(orient(poly[j], poly[i], p)) < EPS:
----------- return True
--- return False
def convex_hull(points):
--- points = sorted(points)
--- def build(h):
------- for p in points:
----------- while len(h) >= 2 and orient(h[-2], h[-1], p) <= EPS:
--------------- h.pop()
----------- h.append(p)
------- return h
--- lower = build([])
--- upper = build([])
--- return lower[:-1] + upper[:-1]
def convex_polygon_inter(a, b):
--- ans = []
--- for p in a:
------- if in_poly(p, b) or on_poly(p, b):
----------- ans.append(p)
--- for p in b:
------- if in_poly(p, a) or on_poly(p, a):
----------- ans.append(p)
--- for i in range(len(a)):
------- I = (i - 1) % len(a)
------- for j in range(len(b)):
----------- J = (j - 1) % len(b)
----------- p = seg_inter(a[I], a[i], b[J], b[j])
----------- if p is not None:
--------------- ans.append(p)
--- if not ans:
------- return []
--- return convex_hull(ans)
```

3.15. Pick Theorem Lattice.

```python
import math

def area(poly):
    n = len(poly)
    s = 0
    for i in range(n):
        j = (i - 1) % n
        s += poly[j][0] * poly[i][1] - poly[i][0] * poly[j][1]
    return abs(s) // 2
def boundary(poly):
    n = len(poly)
    ans = 0
    for i in range(n):
        j = (i - 1) % n
        dx = poly[i][0] - poly[j][0]
        dy = poly[i][1] - poly[j][1]
        ans += math.gcd(abs(dx), abs(dy))
    return ans
def interior(poly):
    A = area(poly)
    B = boundary(poly)
    return A - B // 2 + 1
```

3.16. Minimum Enclosing Circle.

```python
import math

EPS = 1e-9
def dist(a, b):
    return math.hypot(a[0] - b[0], a[1] - b[1])
def mid(a, b):
    return ((a[0] + b[0]) / 2, (a[1] + b[1]) / 2)
def cross(a, b, c):
    return (b[0] - a[0]) * (c[1] - a[1]) - (b[1] - a[1]) * (c[0] - a[0])
def circumcenter(a, b, c):
    ax, ay = a
    bx, by = b
    cx, cy = c
    d = 2 * (ax * (by - cy) + bx * (cy - ay) + cx * (ay - by))
    if abs(d) < EPS:
        return mid(a, b)
    ux = (
        (ax * ax + ay * ay) * (by - cy)
        + (bx * bx + by * by) * (cy - ay)
        + (cx * cx + cy * cy) * (ay - by)
    ) / d
    uy = (
        (ax * ax + ay * ay) * (cx - bx)
        + (bx * bx + by * by) * (ax - cx)
        + (cx * cx + cy * cy) * (bx - ax)
    ) / d
    return (ux, uy)
def bounding_ball(p):
    random.shuffle(p)
    center = p[0]
    radius = 0.0
    n = len(p)
    for i in range(n):
        if dist(center, p[i]) > radius + EPS:
            center = p[i]
            radius = 0.0
            for j in range(i):
                if dist(center, p[j]) > radius + EPS:
                    center = mid(p[i], p[j])
                    radius = dist(center, p[i])
                    for k in range(j):
                        if dist(center, p[k]) > radius + EPS:
                            center = circumcenter(p[i], p[j], p[k])
                            radius = dist(center, p[i])
    return center, radius
def verify(p, center, r):
    for x in p:
        if dist(x, center) > r + 1e-6:
            return False
    return True
```

3.17. Shamos Algorithm.

```python
import math

EPS = 1e-9
def sub(a, b):
    return (a[0] - b[0], a[1] - b[1])
def cross(a, b):
    return a[0] * b[1] - a[1] * b[0]
def dist_pt_line(p, a, b):
    ab = sub(b, a)
    ap = sub(p, a)
    return abs(cross(ab, ap)) / (math.hypot(ab[0], ab[1]) + EPS)
def convex_hull(points):
    points = sorted(points)
    def build():
        hull = []
        for p in points:
            while len(hull) >= 2 and cross(sub(hull[-1], hull[-2]), sub(p, hull[-1])) <= EPS:
                hull.pop()
            hull.append(p)
        return hull
    lower = build()
    upper = build()
    return lower[:-1] + upper[:-1]
def shamos(p):
    h = convex_hull(p)
    k = len(h)

    if k <= 2:
        return 0.0

    d = float("inf")
    j = 1

    for i in range(k):
        ni = (i + 1) % k

        a = h[i]
        b = h[ni]

        while True:
            nj = (j + 1) % k
            if dist_pt_line(h[nj], a, b) > dist_pt_line(h[j], a, b):
                j = nj
            else:
                break
        d = min(d, dist_pt_line(h[j], a, b))
    return d
```

3.18. Kd Tree.

```python
import heapq

EPS = 1e-9
def dist2(a, b):
    return (a[0] - b[0]) ** 2 + (a[1] - b[1]) ** 2
class KDTree:
    def __init__(self, points):
        self.p = points[:]
        self.n = len(points)
        self.build(0, self.n, 0)
    def build(self, L, R, dvx):
        if R - L <= 1:
            return
        mid = (L + R) // 2
        if dvx:
            self.p[L:R] = sorted(self.p[L:R], key=lambda x: x[0])
        else:
            self.p[L:R] = sorted(self.p[L:R], key=lambda x: x[1])
        self.build(L, mid, not dvx)
        self.build(mid + 1, R, not dvx)
    def knn(self, x, y, k=1, r=-1):
        self.qx = x
        self.qy = y
        self.k = k
        self.prune = float("inf") if r < 0 else r * r
        self.heap = []
        self._dfs(0, self.n, False)
        res = [p for _, p in sorted(self.heap, reverse=True)]
        return res
    def _dfs(self, L, R, dvx):
        if R - L <= 0:
            return
        mid = (L + R) // 2
        p = self.p[mid]
        dx = self.qx - p[0]
        dy = self.qy - p[1]
        D = dx * dx + dy * dy
        if D <= self.prune:
            heapq.heappush(self.heap, (-D, p))
            if len(self.heap) > self.k:
                heapq.heappop(self.heap)
        delta = dx if dvx else dy
        nL, nR = L, mid
        fL, fR = mid + 1, R
        if delta > 0:
            nL, fL = fL, nL
            nR, fR = fR, nR
        self._dfs(nL, nR, not dvx)
        if len(self.heap) < self.k or delta * delta <= self.prune:
            self._dfs(fL, fR, not dvx)
```

4. Graphs

4.1. Max Flow.

4.1.1. *Dinic.*

```python
INF = float('inf')
class Dinic:
    def __init__(self, n, s, t):
        self.n, self.s, self.t = n, s, t
        self.edges = []
        self.adj = [[] for _ in range(n)]

    def add_edge(self, u, v, c, bi=False):
        self.adj[u].append(len(self.edges))
        self.edges.append([v, c, 0])
        self.adj[v].append(len(self.edges))
        self.edges.append([u, c if bi else 0, 0])

    def bfs(self):
        self.dist = [-1] * self.n
        self.dist[self.s] = 0
        q = deque([self.s])
        while q:
            u = q.popleft()
            for i in self.adj[u]:
                v, c, f = self.edges[i]
                if self.dist[v] < 0 and c > f:
                    self.dist[v] = self.dist[u] + 1
                    q.append(v)
        return self.dist[self.t] != -1

    def dfs(self, u, pushed):
        if pushed == 0 or u == self.t: return pushed
        while self.ptr[u] < len(self.adj[u]):
```

```
----------- i = self.adj[u][self.ptr[u]] ----------------------------
----------- v, c, f = self.edges[i] ---------------------------------
----------- if self.dist[u]+1 == self.dist[v] and c > f: -------------
--------------- tr = self.dfs(v, min(pushed, c - f)) ------------------
--------------- if tr > 0: --------------------------------------------
------------------- self.edges[i][2] += tr ------------------------
------------------- self.edges[i^1][2] -= tr ----------------------
------------------- return tr -------------------------------------
---------- self.ptr[u] += 1 ---------------------------------------
------- return 0 ---------------------------------------------------
 ------------------------------------------------------------------
--- def calc_max_flow(self): ------------------------------------
------- flow = 0 ---------------------------------------------------
------- while self.bfs(): ------------------------------------------
----------- self.ptr = [0] * self.n ------------------------------
----------- while True: ------------------------------------------
--------------- pushed = self.dfs(self.s, INF) -----------------
--------------- if not pushed: ---------------------------------
------------------- break -----------------------------------
--------------- flow += pushed ----------------------------------
------- return flow ------------------------------------------
```

4.1.2. *Edmonds Karp.*

```
INF = float('inf') ------------------------------------------------
class EdmondsKarp: ------------------------------------------------
--- def __init__(self, n, s, t): ------------------------------
------- self.n, self.s, self.t = n, s, t ---------------------
------- self.adj = [[] for _ in range(n)] --------------------
------- self.res = [[0] * n for _ in range(n)] ---------------
 ------------------------------------------------------------
--- def add_edge(self, u, v, w): ------------------------------
------- self.adj[u].append(v) --------------------------------
------- self.adj[v].append(u) --------------------------------
------- self.res[u][v] += w ----------------------------------
 ------------------------------------------------------------
--- def bfs(self): --------------------------------------------
------- self.par = [-1] * self.n -----------------------------
------- self.par[self.s] = self.s ----------------------------
------- q = deque([self.s]) ----------------------------------
------- while q: ---------------------------------------------
----------- u = q.popleft() --------------------------------
----------- for v in self.adj[u]: --------------------------
--------------- if self.res[u][v] > 0 and self.par[v] == -1: ---------
------------------- self.par[v] = u -------------------------
------------------- if v == self.t: -------------------------
----------------------- return True -----------------------
------------------- q.append(v) -----------------------------
------- return False -----------------------------------------
 ------------------------------------------------------------
--- def calc_max_flow(self): ----------------------------------
------- ans = 0 ----------------------------------------------
------- while self.bfs(): ------------------------------------
----------- flow = INF -------------------------------------
----------- u = self.t -------------------------------------
----------- while u != self.s: -----------------------------
--------------- p = self.par[u] ---------------------------
--------------- flow = min(flow, self.res[p][u]) ----------
--------------- u = p -------------------------------------
----------- u = self.t -------------------------------------
----------- while u != self.s: -----------------------------
--------------- p = self.par[u] ---------------------------
--------------- self.res[p][u] -= flow --------------------
--------------- self.res[u][p] += flow --------------------
--------------- u = p -------------------------------------
----------- ans += flow ------------------------------------
------- return ans -------------------------------------------
```

4.1.3. *Bipartite Matching.*

```
from collections import deque -------------------------------------
 ------------------------------------------------------------------
INF = float('inf') ------------------------------------------------
class BipartiteGraph: ---------------------------------------------
--- def __init__(self, n, m): ---------------------------------
------- self.n = n -------------------------------------------
------- self.m = m -------------------------------------------
------- self.adj = [[] for _ in range(n)] --------------------
------- self.L = [-1] * n ------------------------------------
------- self.R = [-1] * m ------------------------------------
------- self.dist = [INF] * (n + 1) --------------------------
 ------------------------------------------------------------
--- def add_edge(self, u, v): ---------------------------------
------- self.adj[u].append(v) --------------------------------
 ------------------------------------------------------------
--- def bfs(self): --------------------------------------------
------- queue = deque() --------------------------------------
------- for v in range(self.n): ------------------------------
----------- if self.L[v] == -1: ---------------------------
--------------- self.dist[v] = 0 --------------------------
--------------- queue.append(v) ---------------------------
----------- else: -----------------------------------------
--------------- self.dist[v] = INF ------------------------
 ------------------------------------------------------------
------- self.dist[self.n] = INF ------------------------------
 ------------------------------------------------------------
------- while queue: -----------------------------------------
----------- v = queue.popleft() ---------------------------
----------- if self.dist[v] < self.dist[self.n]: ----------
--------------- for u in self.adj[v]: --------------------
------------------- next_v = self.R[u] ----------------
------------------- dist_next = self.dist[next_v] if next_v != -1 else
↪  self.dist[self.n]
 ------------------------------------------------------------
------------------- if (next_v == -1 and self.dist[self.n] == INF) or \
------------------------- (next_v != -1 and self.dist[next_v] ==
↪  INF):
 ------------------------------------------------------------
----------------------- target = next_v if next_v != -1 else self.n --
----------------------- self.dist[target] = self.dist[v] + 1 ---------
----------------------- if target != self.n: --------------------
------------------------- queue.append(target) ----------------
 ------------------------------------------------------------
------- return self.dist[self.n] != INF ----------------------
 ------------------------------------------------------------
--- def dfs(self, v): -----------------------------------------
------- if v != -1: ------------------------------------------
----------- for u in self.adj[v]: -------------------------
--------------- next_v = self.R[u] ------------------------
--------------- dist_next = self.dist[next_v] if next_v != -1 else
↪  self.dist[self.n]
 ------------------------------------------------------------
--------------- if dist_next == self.dist[v] + 1: ---------------
------------------- if self.dfs(next_v): --------------------
----------------------- self.R[u] = v ---------------------
----------------------- self.L[v] = u ---------------------
----------------------- return True -----------------------
----------- self.dist[v] = INF ----------------------------
----------- return False ----------------------------------
------- return True ------------------------------------------
 ------------------------------------------------------------
--- def maximum_matching(self): -------------------------------
------- matching = 0 -----------------------------------------
------- for u in range(self.n): ------------------------------
----------- for v in self.adj[u]: -------------------------
--------------- if self.R[v] == -1: ----------------------
------------------- self.L[u] = v ---------------------
------------------- self.R[v] = u ---------------------
------------------- matching += 1 ---------------------
------------------- break -----------------------------
 ------------------------------------------------------------
------- while self.bfs(): ------------------------------------
----------- for i in range(self.n): -----------------------
--------------- if self.L[i] == -1 and self.dfs(i): ------
------------------- matching += 1 ---------------------
------- return matching --------------------------------------
# Input Format: g = BipartiteGraph(2, 2) ----------------------
#               g.add_edge(0, 0) -------------------------------
#               g.add_edge(0, 1) -------------------------------
#               g.add_edge(1, 0) -------------------------------
#               print(f"Max Matching: {g.maximum_matching()}") -------
```

4.2. Shortest Path.

4.2.1. *Dijkstra.*

```
import heapq ------------------------------------------------------
INF = float('inf') ------------------------------------------------
def dijkstra(n, adj, s): ------------------------------------------
--- dist = [INF] * n ------------------------------------------
--- dist[s] = 0 -----------------------------------------------
--- pq = [(0, s)] ---------------------------------------------
--- ----------------------------------------------------------
--- while pq: -------------------------------------------------
------- d, u = heapq.heappop(pq) -----------------------------
------- if dist[u] < d: --------------------------------------
----------- continue --------------------------------------
------- for v, w in adj[u]: ----------------------------------
----------- if dist[v] > d + w: ---------------------------
--------------- dist[v] = d + w ---------------------------
--------------- heapq.heappush(pq, (dist[v], v)) ----------
--- return dist -----------------------------------------------
```

4.2.2. *Bellman Ford.*

```
INF = float('inf') ------------------------------------------------
def bellman_ford(n, adj, s): --------------------------------------
--- dist = [INF] * n ------------------------------------------
--- dist[s] = 0 -----------------------------------------------
--- for _ in range(n - 1): ------------------------------------
------- for u in range(n): -----------------------------------
----------- if dist[u] == INF: continue -------------------
----------- for v, w in adj[u]: ---------------------------
--------------- if dist[u] + w < dist[v]: -----------------
------------------- dist[v] = dist[u] + w -------------------
--- for u in range(n): ----------------------------------------
------- if dist[u] == INF: continue --------------------------
------- for v, w in adj[u]: ----------------------------------
----------- if dist[u] + w < dist[v]: ---------------------
--------------- return dist, True -------------------------
--- return dist, False ----------------------------------------
```

4.3. Minimum Spanning Tree.

4.3.1. *Kruskal Min Spanning Tree.*

```
import heapq ------------------------------------------------------
 ------------------------------------------------------------------
class UnionFind: --------------------------------------------------
--- def __init__(self, n): ------------------------------
------- self.parent = list(range(n)) -------------------
------- self.rank = [0] * n ----------------------------
 ------------------------------------------------------------
--- def find(self, x): ----------------------------------
------- while self.parent[x] != x: ---------------------
----------- self.parent[x] = self.parent[self.parent[x]] ----------
----------- x = self.parent[x] ----------------------
------- return x ---------------------------------------
 ------------------------------------------------------------
--- def unite(self, a, b): ------------------------------
------- a = self.find(a) -------------------------------
------- b = self.find(b) -------------------------------
 ------------------------------------------------------------
------- if a == b: -------------------------------------
----------- return False ----------------------------------
 ------------------------------------------------------------
```

```
        if self.rank[a] < self.rank[b]:
            self.parent[a] = b
        else:
            self.parent[b] = a
            if self.rank[a] == self.rank[b]:
                self.rank[a] += 1

        return True

def kruskal(n, edges):
    edges = edges[:]
    heapq.heapify(edges)

    uf = UnionFind(n)
    total_weight = 0
    mst = []

    while edges:
        w, u, v = heapq.heappop(edges)
        if uf.unite(u, v):
            mst.append((u, v, w))
            total_weight += w

            if len(mst) == n - 1:
                break

    return mst, total_weight
```

4.3.2. *Prim Min Spanning Tree.*

```
import heapq

def prim(n, adj, start=0):
    visited = [False] * n
    pq = [(0, start, -1)]

    total_weight = 0
    mst = []

    while pq:
        w, u, parent = heapq.heappop(pq)

        if visited[u]:
            continue

        visited[u] = True

        if parent != -1:
            mst.append((parent, u, w))
            total_weight += w

        for v, weight in adj[u]:
            if not visited[v]:
                heapq.heappush(pq, (weight, v, u))

    return mst, total_weight
```

4.4. **Least Common Ancestor.**

4.4.1. *Binary Lifting Lca.*

```
import math

class BinaryLifting:
    def __init__(self, adj):
        self.n = len(adj)
        self.LOG = math.ceil(math.log2(self.n)) + 1
        self.adj = adj

        self.dep = [0] * self.n
        self.par = [[-1] * self.LOG for _ in range(self.n)]

        self.build(0)

    def dfs(self, u, p):
        self.par[u][0] = p
        for v in self.adj[u]:
            if v != p:
                self.dep[v] = self.dep[u] + 1
                self.dfs(v, u)

    def build(self, root):
        self.dfs(root, root)

        for k in range(1, self.LOG):
            for u in range(self.n):
                self.par[u][k] = self.par[self.par[u][k-1]][k-1]

    def lift(self, u, k):
        for i in range(self.LOG):
            if k & (1 << i):
                u = self.par[u][i]
        return u

    def lca(self, u, v):
        if self.dep[u] < self.dep[v]:
            u, v = v, u

        u = self.lift(u, self.dep[u] - self.dep[v])

        if u == v:
            return u

        for k in reversed(range(self.LOG)):
            if self.par[u][k] != self.par[v][k]:
                u = self.par[u][k]
                v = self.par[v][k]

        return self.par[u][0]
```

4.4.2. *Euler Lca.*

```
class EulerLCA:
    def __init__(self, adj):
        self.n = len(adj)
        self.adj = adj

        self.euler = []
        self.first = [-1] * self.n
        self.depth = [0] * self.n

        self.build(0)

    def dfs(self, u, p, d):
        self.first[u] = len(self.euler)
        self.depth[u] = d
        self.euler.append(u)

        for v in self.adj[u]:
            if v != p:
                self.dfs(v, u, d + 1)
                self.euler.append(u)

    def build(self, root):
        self.dfs(root, root, 0)

        m = len(self.euler)
        LOG = m.bit_length()

        self.st = [[0] * m for _ in range(LOG)]
        self.log = [0] * (m + 1)

        for i in range(2, m + 1):
            self.log[i] = self.log[i // 2] + 1

        for i in range(m):
            self.st[0][i] = self.euler[i]

        for k in range(1, LOG):
            for i in range(m - (1 << k) + 1):
                a = self.st[k-1][i]
                b = self.st[k-1][i + (1 << (k-1))]
                self.st[k][i] = a if self.depth[a] < self.depth[b] else
↪ b

    def lca(self, u, v):
        l, r = self.first[u], self.first[v]
        if l > r:
            l, r = r, l

        k = self.log[r - l + 1]
        a = self.st[k][l]
        b = self.st[k][r - (1 << k) + 1]

        return a if self.depth[a] < self.depth[b] else b
```

4.4.3. *Tarjan Lca.*

```
class DSU:
    def __init__(self, n):
        self.p = list(range(n))

    def find(self, x):
        if self.p[x] != x:
            self.p[x] = self.find(self.p[x])
        return self.p[x]

    def unite(self, a, b):
        self.p[self.find(b)] = self.find(a)


class Tarjan:
    def __init__(self, adj, queries):
        self.adj = adj
        self.queries = queries
        self.n = len(adj)

        self.dsu = DSU(self.n)
        self.vis = [False] * self.n
        self.anc = [0] * self.n
        self.res = {}

    def dfs(self, u, p):
        self.anc[u] = u

        for v in self.adj[u]:
            if v == p:
                continue
            self.dfs(v, u)
            self.dsu.unite(u, v)
            self.anc[self.dsu.find(u)] = u

        self.vis[u] = True

        for v, idx in self.queries[u]:
            if self.vis[v]:
                self.res[idx] = self.anc[self.dsu.find(v)]
```

4.5. Minimum Mean Weight Cycle.

```
def min_mean_cycle(adj):
    n = len(adj)
    INF = float('inf')

    arr = [[INF] * n for _ in range(n + 1)]

    for i in range(n):
        arr[0][i] = 0

    for k in range(1, n + 1):
        for u in range(n):
            if arr[k - 1][u] == INF:
                continue
            for v, w in adj[u]:
                arr[k][v] = min(arr[k][v], arr[k - 1][u] + w)

    result = INF

    for v in range(n):
        if arr[n][v] == INF:
            continue

        max_avg = -INF
        for k in range(n):
            if arr[k][v] == INF:
                continue
            avg = (arr[n][v] - arr[k][v]) / (n - k)
            max_avg = max(max_avg, avg)

        result = min(result, max_avg)

    return result
```

4.6. Strongly Connected Components.

```
class KosarajuGraph:
    def __init__(self, n):
        self.n = n
        self.graph = [[] for _ in range(n)]
        self.rev_graph = [[] for _ in range(n)]
        self.vis = [0] * n
        self.sccs = []

    def add_edge(self, u, v):
        self.graph[u].append(v)
        self.rev_graph[v].append(u)

    def dfs(self, u, adj, container):
        self.vis[u] = 1
        for v in adj[u]:
            if not self.vis[v]:
                self.dfs(v, adj, container)
        container.append(u)

    def kosaraju(self):
        topo = []

        # First pass
        self.vis = [0] * self.n
        for i in range(self.n):
            if not self.vis[i]:
                self.dfs(i, self.graph, topo)

        # Second pass
        self.vis = [0] * self.n
        self.sccs = []

        for u in reversed(topo):
            if not self.vis[u]:
                comp = []
                self.dfs(u, self.rev_graph, comp)
                self.sccs.append(comp)

        return self.sccs
```

4.7. Heavy Light Decomposition.

```
class SegmentTree:
    def __init__(self, n):
        self.n = n
        self.tree = [0] * (4 * n)

    # Example: point update
    def update(self, idx, val, node=1, l=0, r=None):
        if r is None:
            r = self.n - 1
        if l == r:
            self.tree[node] = vala
            return
        mid = (l + r) // 2
        if idx <= mid:
            self.update(idx, val, node * 2, l, mid)
        else:
            self.update(idx, val, node * 2 + 1, mid + 1, r)
        self.tree[node] = self.tree[node * 2] + self.tree[node * 2 + 1]

    # Example: range query
    def query(self, ql, qr, node=1, l=0, r=None):
        if r is None:
            r = self.n - 1
        if qr < l or r < ql:
            return 0
        if ql <= l and r <= qr:
            return self.tree[node]
        mid = (l + r) // 2
        return self.query(ql, qr, node * 2, l, mid) + \
               self.query(ql, qr, node * 2 + 1, mid + 1, r)

class HeavyLightTree:
    def __init__(self, n):
        self.n = n
        self.adj = [[] for _ in range(n)]

        self.par = [-1] * n
        self.heavy = [-1] * n
        self.dep = [0] * n
        self.path_root = [0] * n
        self.pos = [0] * n

        self.segment_tree = SegmentTree(n)
        self.current_pos = 0

    def add_edge(self, u, v):
        self.adj[u].append(v)
        self.adj[v].append(u)

    def dfs(self, u):
        size = 1
        max_subtree_size = 0

        for v in self.adj[u]:
            if v != self.par[u]:
                self.par[v] = u
                self.dep[v] = self.dep[u] + 1
                subtree_size = self.dfs(v)

                if subtree_size > max_subtree_size:
                    max_subtree_size = subtree_size
                    self.heavy[u] = v

                size += subtree_size

        return size

    def decompose(self, u, root):
        self.path_root[u] = root
        self.pos[u] = self.current_pos
        self.current_pos += 1

        if self.heavy[u] != -1:
            self.decompose(self.heavy[u], root)

        for v in self.adj[u]:
            if v != self.par[u] and v != self.heavy[u]:
                self.decompose(v, v)

    def build(self, root=0):
        self.par[root] = root
        self.dep[root] = 0
        self.dfs(root)
        self.decompose(root, root)

    # Query path u -> v
    def query(self, u, v):
        res = 0
        while self.path_root[u] != self.path_root[v]:
            if self.dep[self.path_root[u]] <
↪  self.dep[self.path_root[v]]:
                u, v = v, u
            start = self.path_root[u]
            res += self.segment_tree.query(self.pos[start],
↪  self.pos[u])
            u = self.par[start]

        # Same path
        if self.dep[u] > self.dep[v]:
            u, v = v, u
        res += self.segment_tree.query(self.pos[u], self.pos[v])
        return res

    # Update node value
    def update(self, u, value):
        self.segment_tree.update(self.pos[u], value)
```

4.8. Euler Path/Cycle.

4.8.1. *Directed Euler.*

```
from collections import defaultdict

class DirectedEuler:
    def __init__(self, n):
        self.n = n
        self.adj = defaultdict(list)
        self.indeg = [0]*n
        self.outdeg = [0]*n
        self.edge_count = 0

    def add_edge(self, u, v):
        self.adj[u].append(v)
        self.outdeg[u] += 1
        self.indeg[v] += 1
        self.edge_count += 1

    def find_start_node(self):
        start = -1
        end = -1
        for i in range(self.n):
            if self.outdeg[i] - self.indeg[i] == 1:
                if start != -1:
                    return -1
                start = i
            elif self.indeg[i] - self.outdeg[i] == 1:
                if end != -1:
```

```
-------------------- return -1 ----------------------------------------
---------------- end = i ------------------------------------------------
------------ elif self.indeg[i] != self.outdeg[i]: ----------------------
---------------- return -1 ---------------------------------------------
-------------------------------------------------------------------------
------- if start == -1: ---------------------------------------------
------------ for i in range(self.n): -------------------------------------
---------------- if self.outdeg[i] > 0: --------------------------------
-------------------- return i ----------------------------------------
------- return start ------------------------------------------------
-------------------------------------------------------------------------
--- def get_euler_path(self): -----------------------------------------
------- start = self.find_start_node() ------------------------------
------- if start == -1: ---------------------------------------------
------------ return None -------------------------------------------------
-------------------------------------------------------------------------
------- stack = [start] ---------------------------------------------
------- path = [] ---------------------------------------------------
-------------------------------------------------------------------------
------- local_adj = {u: self.adj[u][:] for u in self.adj} ------------
-------------------------------------------------------------------------
------- while stack: ------------------------------------------------
------------ u = stack[-1] -----------------------------------------------
------------ if u in local_adj and local_adj[u]: -------------------------
---------------- v = local_adj[u].pop() --------------------------------
---------------- stack.append(v) ---------------------------------------
------------ else: -------------------------------------------------------
---------------- path.append(stack.pop()) ------------------------------
-------------------------------------------------------------------------
------- if len(path) != self.edge_count + 1: ------------------------
------------ return None -------------------------------------------------
-------------------------------------------------------------------------
------- return path[::-1] -------------------------------------------
-------------------------------------------------------------------------
-------------------------------------------------------------------------
-------------------------------------------------------------------------
--- class UndirectedEuler: --------------------------------------------
------- def __init__(self, n): --------------------------------------
------------ self.n = n --------------------------------------------------
------------ self.adj = defaultdict(list) --------------------------------
------------ self.edge_count = 0 -----------------------------------------
-------------------------------------------------------------------------
------- def add_edge(self, u, v): -----------------------------------
------------ self.adj[u].append(v) ---------------------------------------
------------ self.adj[v].append(u) ---------------------------------------
------------ self.edge_count += 1 ----------------------------------------
-------------------------------------------------------------------------
------- def is_valid(self): -----------------------------------------
------------ odd = 0 -----------------------------------------------------
------------ for i in range(self.n): -------------------------------------
---------------- if len(self.adj[i]) % 2 != 0: -------------------------
-------------------- odd += 1 ----------------------------------------
------------ return odd == 0 or odd == 2 ---------------------------------
-------------------------------------------------------------------------
------- def get_euler_path(self): -----------------------------------
------------ if not self.is_valid(): -------------------------------------
---------------- return None -------------------------------------------
-------------------------------------------------------------------------
------------ start = 0 ---------------------------------------------------
------------ for i in range(self.n): -------------------------------------
---------------- if len(self.adj[i]) % 2 == 1: -------------------------
-------------------- start = i ---------------------------------------
-------------------- break -------------------------------------------
---------------- if len(self.adj[i]) > 0: ------------------------------
-------------------- start = i ---------------------------------------
-------------------------------------------------------------------------
------------ stack = [start] ---------------------------------------------
------------ path = [] ---------------------------------------------------
-------------------------------------------------------------------------
------------ local_adj = {u: self.adj[u][:] for u in self.adj} --------
-------------------------------------------------------------------------
------------ while stack: ------------------------------------------------
---------------- u = stack[-1] -----------------------------------------
---------------- if u in local_adj and local_adj[u]: -------------------
-------------------- v = local_adj[u].pop() --------------------------
-------------------- local_adj[v].remove(u) --------------------------
-------------------- stack.append(v) ---------------------------------
---------------- else: -------------------------------------------------
-------------------- path.append(stack.pop()) ------------------------
-------------------------------------------------------------------------
------------ if len(path) != self.edge_count + 1: ------------------------
---------------- return None -------------------------------------------
-------------------------------------------------------------------------
------------ return path[::-1] -------------------------------------------
```

4.8.2. *Undirected Euler.*

```
from collections import defaultdict -------------------------------------
-------------------------------------------------------------------------
class UndirectedEuler: --------------------------------------------------
--- def __init__(self, n): --------------------------------------------
------- self.n = n --------------------------------------------------
------- self.adj = defaultdict(list) --------------------------------
------- self.edge_count = 0 -----------------------------------------
-------------------------------------------------------------------------
--- def add_edge(self, u, v): -----------------------------------------
------- self.adj[u].append(v) ---------------------------------------
------- self.adj[v].append(u) ---------------------------------------
------- self.edge_count += 1 ----------------------------------------
-------------------------------------------------------------------------
--- def is_valid(self): -----------------------------------------------
------- odd = 0 -----------------------------------------------------
------- for i in range(self.n): -------------------------------------
------------ if len(self.adj[i]) % 2 != 0: -------------------------
---------------- odd += 1 ----------------------------------------
------- return odd == 0 or odd == 2 ---------------------------------
-------------------------------------------------------------------------
--- def get_euler_path(self): -----------------------------------------
------- if not self.is_valid(): -------------------------------------
------------ return None -------------------------------------------
-------------------------------------------------------------------------
------- start = 0 ---------------------------------------------------
------- for i in range(self.n): -------------------------------------
------------ if len(self.adj[i]) % 2 == 1: -------------------------
---------------- start = i ---------------------------------------
---------------- break -------------------------------------------
------------ if len(self.adj[i]) > 0: ------------------------------
---------------- start = i ---------------------------------------
-------------------------------------------------------------------------
------- stack = [start] ---------------------------------------------
------- path = [] ---------------------------------------------------
-------------------------------------------------------------------------
------- local_adj = {u: self.adj[u][:] for u in self.adj} ------------
-------------------------------------------------------------------------
------- while stack: ------------------------------------------------
------------ u = stack[-1] -----------------------------------------
------------ if u in local_adj and local_adj[u]: -------------------
---------------- v = local_adj[u].pop() --------------------------
---------------- local_adj[v].remove(u) --------------------------
---------------- stack.append(v) ---------------------------------
------------ else: -------------------------------------------------
---------------- path.append(stack.pop()) ------------------------
-------------------------------------------------------------------------
------- if len(path) != self.edge_count + 1: ------------------------
------------ return None -------------------------------------------
-------------------------------------------------------------------------
------- return path[::-1] -------------------------------------------
```

4.9. Tree Isomorphism.

```
from collections import deque -------------------------------------------
-------------------------------------------------------------------------
def bfs(u, adj, n): -----------------------------------------------------
--- vis = [False] * n -------------------------------------------------
--- pre = [-1] * n ----------------------------------------------------
--- q = deque() -------------------------------------------------------
-------------------------------------------------------------------------
--- q.append(u) -------------------------------------------------------
--- vis[u] = True -----------------------------------------------------
-------------------------------------------------------------------------
--- last = u ----------------------------------------------------------
-------------------------------------------------------------------------
--- while q: ----------------------------------------------------------
------- u = q.popleft() ---------------------------------------------
------- last = u ----------------------------------------------------
-------------------------------------------------------------------------
------- for v in adj[u]: --------------------------------------------
------------ if not vis[v]: ----------------------------------------
---------------- vis[v] = True -----------------------------------
---------------- pre[v] = u --------------------------------------
---------------- q.append(v) -------------------------------------
-------------------------------------------------------------------------
--- return last, pre --------------------------------------------------
-------------------------------------------------------------------------
def tree_centers(r, adj, n): --------------------------------------------
--- u, _ = bfs(r, adj, n) ---------------------------------------------
--- v, pre = bfs(u, adj, n) -------------------------------------------
-------------------------------------------------------------------------
--- path = [] ---------------------------------------------------------
--- while v != -1: ----------------------------------------------------
------- path.append(v) ----------------------------------------------
------- v = pre[v] --------------------------------------------------
-------------------------------------------------------------------------
--- size = len(path) --------------------------------------------------
-------------------------------------------------------------------------
--- med = [path[size // 2]] -------------------------------------------
--- if size % 2 == 0: -------------------------------------------------
------- med.append(path[size // 2 - 1]) -----------------------------
-------------------------------------------------------------------------
--- return med --------------------------------------------------------
-------------------------------------------------------------------------
def rootcode(u, adj, p, d, pr): -----------------------------------------
--- children = [] -----------------------------------------------------
--- for v in adj[u]: --------------------------------------------------
------- if v != p: --------------------------------------------------
------------ children.append(rootcode(v, adj, u, (d + 1) % len(pr), pr))
-------------------------------------------------------------------------
--- children.sort() ---------------------------------------------------
-------------------------------------------------------------------------
--- h = len(children) + 1 ---------------------------------------------
--- for x in children: ------------------------------------------------
------- h = h * pr[d] + x -------------------------------------------
-------------------------------------------------------------------------
--- return h ----------------------------------------------------------
-------------------------------------------------------------------------
def treecode(root, adj, n, pr): -----------------------------------------
--- c = tree_centers(root, adj, n) ------------------------------------
-------------------------------------------------------------------------
--- if len(c) == 1: ---------------------------------------------------
------- return (rootcode(c[0], adj, -1, 0, pr) << 1) | 1 ------------
--- else: -------------------------------------------------------------
------- return (rootcode(c[0], adj, -1, 0, pr) * ----------------------
---------------- rootcode(c[1], adj, -1, 0, pr)) << 1 ----------------
-------------------------------------------------------------------------
def isomorphic(r1, adj1, r2, adj2, n1, n2, pr, rooted=False): --------
--- if rooted: --------------------------------------------------------
------- return rootcode(r1, adj1, -1, 0, pr) == rootcode(r2, adj2, -1,
↪  0, pr)
--- return treecode(r1, adj1, n1, pr) == treecode(r2, adj2, n2, pr) --
```

4.10. Centroid Decomposition.

```
import sys --------------------------------------------------------------
sys.setrecursionlimit(10**7) --------------------------------------------
-------------------------------------------------------------------------
INF = 10**9 -------------------------------------------------------------
-------------------------------------------------------------------------
```

```python
class CentroidDecomposition:
    def __init__(self, n):
        self.n = n
        self.adj = [[] for _ in range(n)]

        self.LG = 20
        self.jmp = [[-1] * self.LG for _ in range(n)]
        self.path = [[INF] * self.LG for _ in range(n)]

        self.sz = [0] * n
        self.seph = [0] * n
        self.shortest = [INF] * n
        self.blocked = [False] * n

    def add_edge(self, a, b):
        self.adj[a].append(b)
        self.adj[b].append(a)

    def dfs(self, u, p):
        self.sz[u] = 1
        for v in self.adj[u]:
            if v != p and not self.blocked[v]:
                self.sz[u] += self.dfs(v, u)
        return self.sz[u]

    def find_centroid(self, u, p, n):
        for v in self.adj[u]:
            if v != p and not self.blocked[v] and self.sz[v] > n // 2:
                return self.find_centroid(v, u, n)
        return u

    def makepaths(self, sep, u, p, depth, h):
        self.jmp[u][h] = sep
        self.path[u][h] = depth

        for v in self.adj[u]:
            if v != p and not self.blocked[v]:
                self.makepaths(sep, v, u, depth + 1, h)

    def separate(self, h=0, u=0):
        self.dfs(u, -1)
        c = self.find_centroid(u, -1, self.sz[u])

        self.seph[c] = h
        self.makepaths(c, c, -1, 0, h)

        self.blocked[c] = True

        for v in self.adj[c]:
            if not self.blocked[v]:
                self.separate(h + 1, v)

    def paint(self, u):
        for h in range(self.seph[u] + 1):
            c = self.jmp[u][h]
            self.shortest[c] = min(self.shortest[c], self.path[u][h])

    def closest(self, u):
        ans = INF
        for h in range(self.seph[u] + 1):
            c = self.jmp[u][h]
            ans = min(ans, self.shortest[c] + self.path[u][h])
        return ans
```

5. Math I Algebra

5.0.1. *Generating Function Manager.*

```python
MOD = 998244353

def mod_pow(base, exp):
    res = 1
    base %= MOD
    while exp:
        if exp & 1:
            res = res * base % MOD
        base = base * base % MOD
        exp >>= 1
    return res


class GF_Manager:
    DEPTH = 23

    def __init__(self):
        self.prim = [0] * (self.DEPTH + 1)
        self.prim_inv = [0] * (self.DEPTH + 1)
        self.two_inv = [0] * (self.DEPTH + 1)
        self.set_up_primitives()

    def set_up_primitives(self):
        self.prim[self.DEPTH] = 31
        self.prim_inv[self.DEPTH] = mod_pow(self.prim[self.DEPTH], MOD - 2)
        self.two_inv[self.DEPTH] = mod_pow(1 << self.DEPTH, MOD - 2)

        for n in range(self.DEPTH - 1, -1, -1):
            self.prim[n] = self.prim[n + 1] * self.prim[n + 1] % MOD
            self.prim_inv[n] = mod_pow(self.prim[n], MOD - 2)
            self.two_inv[n] = mod_pow(1 << n, MOD - 2)

    def NTT(self, A, n, is_inverse=False):
        if n == 0:
            return A[:]

        half = 1 << (n - 1)
        even = A[0::2]
        odd = A[1::2]

        even = self.NTT(even, n - 1, is_inverse)
        odd = self.NTT(odd, n - 1, is_inverse)

        res = [0] * (1 << n)
        w1 = self.prim_inv[n] if is_inverse else self.prim[n]
        w = 1

        for i in range(half):
            t = w * odd[i] % MOD
            res[i] = (even[i] + t) % MOD
            res[i + half] = (even[i] - t) % MOD
            w = w * w1 % MOD

        return res

    def mult(self, A, B):
        an, bn = len(A), len(B)
        degree = an + bn - 1

        n = 0
        while (1 << n) < degree:
            n += 1

        size = 1 << n
        tA = A + [0] * (size - an)
        tB = B + [0] * (size - bn)

        tA = self.NTT(tA, n)
        tB = self.NTT(tB, n)

        for i in range(size):
            tA[i] = tA[i] * tB[i] % MOD

        tA = self.NTT(tA, n, True)
        inv = self.two_inv[n]
        return [(x * inv) % MOD for x in tA[:degree]]

    def horners(self, F, x):
        ans = 0
        for coef in reversed(F):
            ans = (ans * x + coef) % MOD
        return ans
```

5.0.2. *Fast Fourier Transform.*

```python
import math

class Poly:
    def __init__(self, a=0.0, b=0.0):
        self.a = a
        self.b = b

    def __add__(self, other):
        return Poly(self.a + other.a, self.b + other.b)

    def __sub__(self, other):
        return Poly(self.a - other.a, self.b - other.b)

    def __mul__(self, other):
        return Poly(
            self.a * other.a - self.b * other.b,
            self.a * other.b + self.b * other.a
        )

    def __repr__(self):
        return f"({self.a}+{self.b}i)"

def fft_recursive(inp, n, s):
    if n == 1:
        return [inp[0]]

    half = n // 2

    even = fft_recursive(inp, half, s * 2)
    odd = fft_recursive(inp[s:], half, s * 2)

    res = [None] * n
    w = Poly(1, 0)
    wn = Poly(math.cos(math.pi / half), math.sin(math.pi / half))

    for i in range(half):
        t = w * odd[i]
        res[i] = even[i] + t
        res[i + half] = even[i] - t
        w = w * wn

    return res

def fft(p):
    return fft_recursive(p, len(p), 1)

def inverse_fft(p):
    n = len(p)

    p = [Poly(x.a, -x.b) for x in p]

    p = fft(p)

    res = []
    for x in p:
        res.append(Poly(x.a / n, -x.b / n))

    return res
```

5.0.3. *Fft Polynomial Multiplication.*

```
def fft_multiply(a, b):
    an, bn = len(a), len(b)
    degree = an + bn - 1

    n = 1
    while n < degree:
        n <<= 1

    A = [Poly(x, 0) for x in a] + [Poly(0, 0)] * (n - an)
    B = [Poly(x, 0) for x in b] + [Poly(0, 0)] * (n - bn)

    A = fft(A)
    B = fft(B)

    for i in range(n):
        A[i] = A[i] * B[i]

    A = inverse_fft(A)

    result = [int(round(A[i].a)) for i in range(degree)]
    return result
```

5.0.4. *Number Theoretic Transform.*

```
MOD = 998244353
PRIMITIVE_ROOT = 3

def mod_pow(base, exp, mod=MOD):
    res = 1
    base %= mod
    while exp:
        if exp & 1:
            res = res * base % mod
        base = base * base % mod
        exp >>= 1
    return res

class Num:
    def __init__(self, x=0):
        self.x = x % MOD

    def __add__(self, other):
        return Num(self.x + other.x)

    def __sub__(self, other):
        return Num(self.x - other.x)

    def __mul__(self, other):
        return Num(self.x * other.x)

    def __truediv__(self, other):
        return self * other.inv()

    def inv(self):
        return Num(mod_pow(self.x, MOD - 2))

    def pow(self, p):
        return Num(mod_pow(self.x, p))

    def __repr__(self):
        return str(self.x)

def ntt(a, invert=False):
    n = len(a)

    j = 0
    for i in range(1, n):
        bit = n >> 1
        while j & bit:
            j ^= bit
            bit >>= 1
        j |= bit
        if i < j:
            a[i], a[j] = a[j], a[i]

    length = 2
    while length <= n:
        wlen = mod_pow(PRIMITIVE_ROOT, (MOD - 1) // length)
        if invert:
            wlen = mod_pow(wlen, MOD - 2)

        for i in range(0, n, length):
            w = 1
            half = length // 2
            for j in range(i, i + half):
                u = a[j]
                v = Num(a[j + half].x * w % MOD)
                a[j] = u + v
                a[j + half] = u - v
                w = w * wlen % MOD

        length <<= 1

    if invert:
        inv_n = mod_pow(n, MOD - 2)
        for i in range(n):
            a[i] = Num(a[i].x * inv_n % MOD)

def multiply(A, B):
    n = 1
    while n < len(A) + len(B):
        n <<= 1

    fa = [Num(x) for x in A] + [Num(0)] * (n - len(A))
    fb = [Num(x) for x in B] + [Num(0)] * (n - len(B))

    ntt(fa)
    ntt(fb)

    for i in range(n):
        fa[i] = fa[i] * fb[i]

    ntt(fa, True)

    return [fa[i].x for i in range(len(A) + len(B) - 1)]

def poly_inv(x, l):
    if l == 1:
        return [mod_pow(x[0], MOD - 2)]

    y = poly_inv(x, (l + 1) // 2)

    n = 1
    while n < 2 * l:
        n <<= 1

    A = x[:l] + [0] * (n - l)
    B = y[:] + [0] * (n - len(y))

    A = [Num(v) for v in A]
    B = [Num(v) for v in B]

    ntt(A)
    ntt(B)

    for i in range(n):
        B[i] = B[i] * (Num(2) - A[i] * B[i])

    ntt(B, True)

    return [B[i].x % MOD for i in range(l)]

def poly_sqrt(x, l):
    if l == 1:
        assert x[0] == 1
        return [1]

    y = poly_sqrt(x, l // 2)
    inv_y = poly_inv(y, l // 2)

    n = 1
    while n < 2 * l:
        n <<= 1

    T1 = x[:l] + [0] * (n - l)
    T2 = inv_y[:] + [0] * (n - len(inv_y))

    T1 = [Num(v) for v in T1]
    T2 = [Num(v) for v in T2]

    ntt(T1)
    ntt(T2)

    for i in range(n):
        T2[i] = T1[i] * T2[i]

    ntt(T2, True)

    inv2 = (MOD + 1) // 2
    res = []
    for i in range(l):
        val = (y[i] + T2[i].x) * inv2 % MOD
        res.append(val)

    return res
```

5.0.5. *Polynomial Long Division.*

```
MOD = 998244353
G = 3

def mod_pow(a, e):
    r = 1
    while e:
        if e & 1:
            r = r * a % MOD
        a = a * a % MOD
        e >>= 1
    return r

def bit_reverse(a):
    n = len(a)
    j = 0
    for i in range(1, n):
        bit = n >> 1
        while j & bit:
            j ^= bit
            bit >>= 1
        j ^= bit
        if i < j:
            a[i], a[j] = a[j], a[i]

def ntt(a, invert=False):
    n = len(a)
    bit_reverse(a)

    length = 2
    while length <= n:
        wlen = mod_pow(G, (MOD - 1) // length)
        if invert:
            wlen = mod_pow(wlen, MOD - 2)

        for i in range(0, n, length):
            w = 1
            half = length // 2
```

```
---------- for j in range(i, i + half):
---------- u = a[j]
---------- v = a[j + half] * w % MOD
---------- a[j] = (u + v) % MOD
---------- a[j + half] = (u - v) % MOD
---------- w = w * wlen % MOD

------- length <<= 1

--- if invert:
------- inv_n = mod_pow(n, MOD - 2)
------- for i in range(n):
---------- a[i] = a[i] * inv_n % MOD

def multiply(a, b):
--- n = 1
--- while n < len(a) + len(b):
------- n <<= 1

--- fa = a[:] + [0] * (n - len(a))
--- fb = b[:] + [0] * (n - len(b))

--- ntt(fa)
--- ntt(fb)

--- for i in range(n):
------- fa[i] = fa[i] * fb[i] % MOD

--- ntt(fa, True)

--- return fa[:len(a) + len(b) - 1]

def poly_inv(f, n):
--- if n == 1:
------- return [mod_pow(f[0], MOD - 2)]

--- m = (n + 1) // 2
--- g = poly_inv(f, m)

--- size = 1
--- while size < 2 * n:
------- size <<= 1

--- a = f[:n] + [0] * (size - len(f[:n]))
--- g = g + [0] * (size - len(g))

--- ntt(a)
--- ntt(g)

--- for i in range(size):
------- g[i] = g[i] * (2 - a[i] * g[i] % MOD) % MOD

--- ntt(g, True)

--- return g[:n]

def poly_divide(A, B):
--- n, m = len(A), len(B)

--- if n < m:
------- return [], A[:]

--- k = n - m + 1

--- A_rev = A[::-1]
--- B_rev = B[::-1]

--- B_inv = poly_inv(B_rev, k)

--- Q_rev = multiply(A_rev, B_inv)[:k]
--- Q = Q_rev[::-1]

--- BQ = multiply(B, Q)

--- R = [(A[i] - (BQ[i] if i < len(BQ) else 0)) % MOD for i in
↪  range(n)]

--- while Q and Q[-1] == 0:
------- Q.pop()
--- while R and R[-1] == 0:
------- R.pop()

--- return Q, R
```

5.0.6. *Matrix Multiplication.*

```
MOD = 998244353

def multiply(A, B):
--- p = len(A)
--- q = len(A[0])
--- r = len(B[0])

--- if len(B) != q:
------- raise ValueError("Dimension mismatch")

--- AB = [[0] * r for _ in range(p)]

--- for i in range(p):
------- for j in range(q):
---------- for k in range(r):
---------- AB[i][k] = (AB[i][k] + A[i][j] * B[j][k]) % MOD

--- return AB
```

5.0.7. *Matrix Power.*

```
MOD = 998244353

def multiply(A, B):
--- p, q, r = len(A), len(A[0]), len(B[0])

--- res = [[0] * r for _ in range(p)]

--- for i in range(p):
------- for j in range(q):
---------- for k in range(r):
---------- res[i][k] = (res[i][k] + A[i][j] * B[j][k]) % MOD

--- return res

def identity(n):
--- I = [[0] * n for _ in range(n)]
--- for i in range(n):
------- I[i][i] = 1
--- return I

def power(B, e):
--- n = len(B)

--- result = identity(n)
--- base = [row[:] for row in B]

--- while e > 0:
------- if e & 1:
---------- result = multiply(result, base)
------- base = multiply(base, base)
------- e >>= 1

--- return result
```

5.0.8. *Gauss Jordan Matrix Determinant.*

```
EPS = 1e-9

def gauss_jordan(A):
--- n = len(A)
--- m = len(A[0])

--- singular = False

--- i = 0
--- for p in range(m):
------- if i >= n:
---------- break

------- pivot = i
------- for k in range(i + 1, n):
---------- if abs(A[k][p]) > abs(A[pivot][p]):
---------- pivot = k

------- if abs(A[pivot][p]) < EPS:
---------- singular = True
---------- continue

------- A[i], A[pivot] = A[pivot], A[i]

------- div = A[i][p]
------- for j in range(p, m):
---------- A[i][j] /= div

------- for k in range(n):
---------- if k == i:
---------- continue
---------- factor = A[k][p]
---------- for j in range(p, m):
---------- A[k][j] -= factor * A[i][j]

------- i += 1

--- return not singular

def apply(A, x):
--- return [sum(A[i][j] * x[j] for j in range(len(x))) for i in
↪  range(len(A))]
```

6. Math II Combinatorics

6.0.1. *Lucas Theorem.*

```
_fact_cache = []
_last_p = -1

def lucas(n, k, p):
--- if k < 0 or k > n:
------- return 0
--- if k == 0 or n == k:
------- return 1

--- global _fact_cache, _last_p
--- if _last_p != p:
------- _fact_cache = [1] * p
------- for i in range(2, p):
---------- _fact_cache[i] = (_fact_cache[i - 1] * i) % p
------- _last_p = p

--- res_num = 1
--- res_den = 1

--- while n > 0 or k > 0:
------- ni, ki = n % p, k % p
------- if ki > ni:
---------- return 0
------- res_num = (res_num * _fact_cache[ni]) % p
```

```
------- res_den = (res_den * _fact_cache[ki] * _fact_cache[ni - ki]) %
↪  p
------- n //= p
------- k //= p
--- return (res_num * pow(res_den, p - 2, p)) % p
# Input Format: print(lucas(n, k, p))
```

6.0.2. *Catalan Numbers.*

```
def catalan(n):
--- if (n == 0 or n == 1):
------- return 1
---
--- dp = [0] * (n + 1)
--- dp[0] = dp[1] = 1

--- for i in range(2, n + 1):
------- for j in range(i):
----------- dp[i] += dp[j] * dp[i-j-1]

--- return dp[n]

# for i in range(10):
#     print(catalan(i))
```

6.0.3. *Kth Permutation.*

```
#fact = [1, 1, 2, 6, 24, 120, 720, 5040, 40320, 362880]

def findKthPermutation(n, k):
--- nums = [str(i) for i in range(1, n + 1)]

--- fact = [1] * n
--- for i in range(1, n):
------- fact[i] = fact[i - 1] * i

--- k -= 1
--- res = []

--- for i in range(n - 1, -1, -1):
------- idx = k // fact[i]
------- k %= fact[i]

------- res.append(nums.pop(idx))

--- return ''.join(res)
```

6.0.4. *Factoradics.*

```
def factoradic(arr):
--- n = len(arr)
--- ft = FenwickTree(n)
--- for i in range(1, n + 1):
------- ft.add(i, 1)

--- res = []
--- for val in arr:
------- bit_val = val + 1
------- s = ft.query(bit_val - 1)
------- ft.add(bit_val, -1)
------- res.append(s)
--- return res

def permute(fact, n):
--- ft = FenwickTree(n)
--- for i in range(1, n + 1):
------- ft.add(i, 1)

--- res = []
--- for val in fact:
------- idx = ft.find_kth(val)
------- ft.add(idx, -1)
------- res.append(idx - 1)
--- return res
```

7. Math III Number Theory

7.0.1. *Gcd Subset Counter.*

```
def gcd_subset_count(arr):
--- n = len(arr)
--- mx = max(arr)

--- f = [0] * (mx + 1)
--- for x in arr:
------- f[x] += 1

--- gcnt = [0] * (mx + 1)

--- for i in range(mx, 0, -1):
------- add = 0

------- for j in range(i, mx + 1, i):
----------- add += f[j]

------- total = (1 << add) - 1  # 2^add - 1

------- sub = 0
------- for j in range(2 * i, mx + 1, i):
----------- sub += gcnt[j]

------- gcnt[i] = total - sub

--- return gcnt
```

7.0.2. *Linear Diophantine.*

```
def gcd(a, b):
--- if a == 0:
------- return b
--- return gcd(b % a, a)

def isPossible(a, b, c):
--- g = gcd(a, b)
---
--- return c % g == 0
```

7.0.3. *Chinese Remainder Theorem.*

```
def gcd_extended(a, b):
--- if a == 0:
------- return b, 0, 1
--- gcd, x1, y1 = gcd_extended(b % a, a)
--- x = y1 - (b // a) * x1
--- y = x1
--- return gcd, x, y

def find_min_x(num, rem):
--- prod = 1
--- for n in num:
------- prod *= n

--- result = 0
--- for i in range(len(num)):
------- prod_i = prod // num[i]
------- _, inv_i, _ = gcd_extended(prod_i, num[i])
------- result += rem[i] * prod_i * inv_i

--- return result % prod
```

7.0.4. *Euler Totient.*

```
def totient(n):
--- if n <= 1:
------- return 1

--- tot = n
--- i = 2

--- while i * i <= n:
------- if n % i == 0:
----------- tot -= tot // i
----------- while n % i == 0:
--------------- n //= i
------- i += 1

--- if n > 1:
------- tot -= tot // n

--- return tot
```

7.0.5. *Extended Euclidean Algorithm.*

```
def mod(x, m):
--- if m == 0:
------- return 0
--- if m < 0:
------- m = -m
--- return (x % m + m) % m

def extended_euclid(a, b):
--- if b == 0:
------- return a, 1, 0

--- g, x1, y1 = extended_euclid(b, a % b)

--- x = y1
--- y = x1 - (a // b) * y1

--- return g, x, y
```

7.0.6. *Modular Exponentiation.*

```
def mod_pow(b, e, m):
--- res = 1 % m

--- while e > 0:
------- if e & 1:
----------- res = (res * b) % m
------- b = (b * b) % m
------- e >>= 1

--- return res
```

7.0.7. *Mod Inverse.*

```
def extended_euclid(a, b):
--- if b == 0:
------- return a, 1, 0

--- g, x1, y1 = extended_euclid(b, a % b)

--- x = y1
--- y = x1 - (a // b) * y1

--- return g, x, y

def modinv(a, m):
--- g, x, y = extended_euclid(a, m)

--- if g != 1:
------- return 0  # no inverse

--- return x % m
```

7.0.8. *Modulo Solver.*

```
def extended_euclid(a, b):
    if b == 0:
        return a, 1, 0

    g, x1, y1 = extended_euclid(b, a % b)

    x = y1
    y = x1 - (a // b) * y1

    return g, x, y


def modsolver(a, b, m):
    g, x, y = extended_euclid(a, m)

    if b % g != 0:
        return (-1, -1)

    m_reduced = m // g

    x0 = (x * (b // g)) % m_reduced

    return (x0, m_reduced)
```

7.0.9. *Primitive Root.*

```
def primitive_root(m):
    if m == 2:
        return 1

    phi = m - 1

    factors = set()
    x = phi
    i = 2

    while i * i <= x:
        if x % i == 0:
            factors.add(i)
            while x % i == 0:
                x //= i
        i += 1

    if x > 1:
        factors.add(x)

    for g in range(2, m):
        ok = True
        for f in factors:
            if mod_pow(g, phi // f, m) == 1:
                ok = False
                break
        if ok:
            return g

    return -1
```

7.0.10. *Josephus.*

```
def josephus(n, k):
    res = 0
    for i in range(2, n + 1):
        res = (res + k) % i
    return res
```

8. Math IV Numerical Methods

8.0.1. *Fast Square Testing.*

```
import math

class FastSquareTester:
    def __init__(self):
        self.M = 0
        self.init_is_square()

    def init_is_square(self):
        for i in range(64):
            self.M |= (1 << (i * i % 64))

    def is_square(self, x):
        if x < 0: return False
        if x == 0: return True

        if not (self.M & (1 << (x & 63))):
            return False

        c = (x & -x).bit_length() - 1
        if c & 1:
            return False

        temp_x = x >> c
        if (temp_x & 7) != 1:
            return False

        r = int(math.isqrt(temp_x))
        return r * r == temp_x

tester = FastSquareTester()
```

8.0.2. *Simpson Integration.*

```
def f(x):
    return x ** 2
def simpson_integration(a, b, n=1000000):
    if n % 2 != 0:
        n += 1

    h = (b - a) / n
    s = f(a) + f(b)

    for i in range(1, n):
        x = a + h * i
        if i % 2 == 1:
            s += 4 * f(x)
        else:
            s += 2 * f(x)

    return s * (h / 3)
```

9. Strings

9.0.1. *Longest Common Prefix.*

```
def build_lcp(s, sa):
    n = len(s)
    rank = [0] * n
    lcp = [0] * n

    for i in range(n):
        rank[sa[i]] = i

    k = 0
    for i in range(n):
        if rank[i] == 0:
            k = 0
            continue

        j = sa[rank[i] - 1]

        while i + k < n and j + k < n and s[i + k] == s[j + k]:
            k += 1

        lcp[rank[i]] = k

        if k > 0:
            k -= 1

    return lcp
```

9.0.2. *Hasher.*

```
class Hasher:
    def __init__(self, s, bases=[31, 37], mod=10 ** 9 + 7):
        self.bases = bases
        self.mod = mod
        self.n = len(s)
        self.num_hashes = len(bases)
        self.p_pow = [[1] * (self.n + 1) for _ in range(self.num_hashes)]
        self.h_ans = [[0] * (self.n + 1) for _ in range(self.num_hashes)]

        for i, b in enumerate(self.bases):
            for j in range(self.n):
                self.p_pow[i][j + 1] = (self.p_pow[i][j] * b) % self.mod
                self.h_ans[i][j + 1] = (self.h_ans[i][j] + s[j] * self.p_pow[i][j]) % self.mod

    def get_hash(self, l, r):
        return tuple((self.h_ans[i][r + 1] - self.h_ans[i][l]) % self.mod
                     for i in range(self.num_hashes))

    def are_equal(self, l1, r1, l2, r2):
        if (r1 - l1) != (r2 - l2): return False

        h1 = self.get_hash(l1, r1)
        h2 = self.get_hash(l2, r2)

        return all((h1[i] * self.p_pow[i][l2]) % self.mod == (h2[i] * self.p_pow[i][l1]) % self.mod
                   for i in range(self.num_hashes))
```

9.0.3. *Z Algorithm.*

9.0.4. *Aho Corasick Trie.*

```
from collections import deque
import random
import string

class Node:
    def __init__(self):
        self.next = {}
        self.fail = None
        self.count = 0

    def add(self, s):
        node = self
        for c in s:
            if c not in node.next:
                node.next[c] = Node()
            node = node.next[c]
        node.count += 1

    def prepare(self):
        root = self
        q = deque()

        root.fail = root
```

```
-------------------------------------------------------------------------
------- for child in root.next.values(): ------------------------------
----------- child.fail = root -------------------------------------------
----------- q.append(child) ---------------------------------------------
-------------------------------------------------------------------------
------- while q: ------------------------------------------------------
----------- head = q.popleft() ------------------------------------------
-------------------------------------------------------------------------
----------- for c, nxt in head.next.items(): ----------------------------
--------------- f = head.fail -----------------------------------------
-------------------------------------------------------------------------
--------------- while f != root and c not in f.next: ------------------
------------------- f = f.fail ----------------------------------------
-------------------------------------------------------------------------
--------------- nxt.fail = f.next[c] if c in f.next else root --------
-------------------------------------------------------------------------
--------------- nxt.count += nxt.fail.count ---------------------------
-------------------------------------------------------------------------
--------------- q.append(nxt) -----------------------------------------
-------------------------------------------------------------------------
--- def search(self, s): ------------------------------------------------
------- root = self ---------------------------------------------------
------- p = self ------------------------------------------------------
------- ans = 0 -------------------------------------------------------
-------------------------------------------------------------------------
------- for c in s: ---------------------------------------------------
----------- while p != root and c not in p.next: ----------------------
--------------- p = p.fail --------------------------------------------
-------------------------------------------------------------------------
----------- if c in p.next: ---------------------------------------------
--------------- p = p.next[c] -----------------------------------------
--------------- ans += p.count ----------------------------------------
-------------------------------------------------------------------------
------- return ans ----------------------------------------------------
```

9.0.5. *Knuth Morris Pratt.*

```
def build_kmp(f): ---------------------------------------------------
--- n = len(f) ----------------------------------------------------
--- par = [0] * (n + 1) -------------------------------------------
-------------------------------------------------------------------
--- par[0] = -1 ---------------------------------------------------
--- if n > 0: -----------------------------------------------------
------- par[1] = 0 ----------------------------------------------
-------------------------------------------------------------------
--- i, j = 2, 0 ---------------------------------------------------
-------------------------------------------------------------------
--- while i <= n: -------------------------------------------------
------- if f[i - 1] == f[j]: ------------------------------------
----------- j += 1 --------------------------------------------
----------- par[i] = j ----------------------------------------
----------- i += 1 --------------------------------------------
------- elif j > 0: ---------------------------------------------
----------- j = par[j] ----------------------------------------
------- else: ---------------------------------------------------
----------- par[i] = 0 ----------------------------------------
----------- i += 1 --------------------------------------------
-------------------------------------------------------------------
--- return par ----------------------------------------------------
-------------------------------------------------------------------
-------------------------------------------------------------------
def kmp(s, f): ------------------------------------------------------
--- par = build_kmp(f) --------------------------------------------
-------------------------------------------------------------------
--- i = 0 ---------------------------------------------------------
--- j = 0 ---------------------------------------------------------
--- ans = [] ------------------------------------------------------
-------------------------------------------------------------------
--- n, m = len(s), len(f) -----------------------------------------
-------------------------------------------------------------------
--- while i + j < n: ----------------------------------------------
------- if s[i + j] == f[j]: ------------------------------------
----------- j += 1 --------------------------------------------
----------- if j == m: ----------------------------------------
--------------- ans.append(i) -----------------------------
--------------- i += j - par[j] ---------------------------
--------------- if j > 0: ---------------------------------
------------------- j = par[j] --------------------------
------- else: ---------------------------------------------
----------- i += j - par[j] -----------------------------
----------- if j > 0: -----------------------------------
--------------- j = par[j] --------------------------------
-------------------------------------------------------------
--- return ans ----------------------------------------------------
```

9.0.6. *Trie.*

```
class Node: ---------------------------------------------------------
--- def __init__(self): -------------------------------------------
------- self.children = {} --------------------------------------
------- self.prefixes = 0 ---------------------------------------
------- self.words = 0 ------------------------------------------
-------------------------------------------------------------------
class Trie: ---------------------------------------------------------
--- def __init__(self): -------------------------------------------
------- self.root = Node() --------------------------------------
-------------------------------------------------------------------
--- def insert(self, seq): ----------------------------------------
------- cur = self.root -----------------------------------------
-------------------------------------------------------------------
------- for i in range(len(seq) + 1): ---------------------------
----------- cur.prefixes += 1 ---------------------------------
-------------------------------------------------------------------
----------- if i == len(seq): ---------------------------------
--------------- cur.words += 1 ----------------------------------
--------------- break -------------------------------------------
-------------------------------------------------------------------
----------- head = seq[i] -------------------------------------
-------------------------------------------------------------------
----------- if head not in cur.children: ----------------------
--------------- cur.children[head] = Node() ---------------------
-------------------------------------------------------------------
----------- cur = cur.children[head] --------------------------
-------------------------------------------------------------------
--- def countMatches(self, seq): ----------------------------------
------- cur = self.root -----------------------------------------
-------------------------------------------------------------------
------- for c in seq: -------------------------------------------
----------- if c not in cur.children: -------------------------
--------------- return 0 ----------------------------------------
----------- cur = cur.children[c] -----------------------------
-------------------------------------------------------------------
------- return cur.words ----------------------------------------
-------------------------------------------------------------------
--- def countPrefixes(self, seq): ---------------------------------
------- cur = self.root -----------------------------------------
-------------------------------------------------------------------
------- for c in seq: -------------------------------------------
----------- if c not in cur.children: -------------------------
--------------- return 0 ----------------------------------------
----------- cur = cur.children[c] -----------------------------
-------------------------------------------------------------------
------- return cur.prefixes -------------------------------------
```

9.0.7. *Suffix Array.*

```
class SuffixArray: --------------------------------------------------
--- def __init__(self): -------------------------------------------
------- self.n = 0 ----------------------------------------------
------- self.suffix = [] ----------------------------------------
------- self.equiv = [] -----------------------------------------
------- self.equiv_pair = [] ------------------------------------
------- self.s = "" ---------------------------------------------
-------------------------------------------------------------------
--- def build(self, s): -------------------------------------------
------- if not s.endswith('$'): ---------------------------------
----------- s += '$' ------------------------------------------
-------------------------------------------------------------------
------- self.s = s ----------------------------------------------
------- self.n = len(s) -----------------------------------------
-------------------------------------------------------------------
------- self.suffix = list(range(self.n)) -----------------------
------- self.equiv = [0] * self.n -------------------------------
------- self.equiv_pair = [(0, 0)] * self.n ---------------------
-------------------------------------------------------------------
------- self.suffix.sort(key=lambda i: s[i]) --------------------
-------------------------------------------------------------------
------- cls = 0 -------------------------------------------------
------- for i in range(self.n): ---------------------------------
----------- if i == 0 or s[self.suffix[i]] != s[self.suffix[i - 1]]: -
--------------- cls += 1 ----------------------------------------
----------- self.equiv[self.suffix[i]] = cls ------------------
-------------------------------------------------------------------
------- k = 1 ---------------------------------------------------
------- while k < self.n: ---------------------------------------
----------- for i in range(self.n): ---------------------------
--------------- self.equiv_pair[i] = ( --------------------------
------------------- self.equiv[i], ----------------------------
------------------- self.equiv[(i + k) % self.n] --------------
--------------- ) -----------------------------------------------
-------------------------------------------------------------------
----------- self.suffix.sort(key=lambda i: self.equiv_pair[i]) ------
-------------------------------------------------------------------
----------- cls = 0 -------------------------------------------
----------- new_equiv = [0] * self.n --------------------------
-------------------------------------------------------------------
----------- for i in range(self.n): ---------------------------
--------------- if i == 0 or self.equiv_pair[self.suffix[i]] !=
↪  self.equiv_pair[self.suffix[i - 1]]:
------------------- cls += 1 ----------------------------------
--------------- new_equiv[self.suffix[i]] = cls -----------------
-------------------------------------------------------------------
----------- self.equiv = new_equiv ----------------------------
----------- k <<= 1 -------------------------------------------
-------------------------------------------------------------------
--- def _cmp(self, start, pattern): -------------------------------
------- s = self.s ----------------------------------------------
------- n = len(s) ----------------------------------------------
-------------------------------------------------------------------
------- i = start -----------------------------------------------
------- j = 0 ---------------------------------------------------
-------------------------------------------------------------------
------- while i < n and j < len(pattern): -----------------------
----------- if s[i] != pattern[j]: ----------------------------
--------------- return ord(s[i]) - ord(pattern[j]) --------------
----------- i += 1 --------------------------------------------
----------- j += 1 --------------------------------------------
-------------------------------------------------------------------
------- if j == len(pattern): -----------------------------------
----------- return 0 ------------------------------------------
------- return -1 -----------------------------------------------
-------------------------------------------------------------------
--- def lower_bound(self, pattern): -------------------------------
------- l, r = 0, self.n - 1 ------------------------------------
------- res = self.n --------------------------------------------
-------------------------------------------------------------------
------- while l <= r: -------------------------------------------
----------- m = (l + r) // 2 ----------------------------------
----------- if self._cmp(self.suffix[m], pattern) >= 0: -------
--------------- res = m -----------------------------------------
--------------- r = m - 1 ---------------------------------------
----------- else: ---------------------------------------------
--------------- l = m + 1 ---------------------------------------
-------------------------------------------------------------------
------- return res ----------------------------------------------
-------------------------------------------------------------------
--- def upper_bound(self, pattern): -------------------------------
```

```
------- l, r = 0, self.n - 1
------- res = self.n
------- while l <= r:
----------- m = (l + r) // 2
----------- if self._cmp(self.suffix[m], pattern) > 0:
--------------- res = m
--------------- r = m - 1
----------- else:
--------------- l = m + 1
------- return res
--- def count_occurrences(self, pattern):
------- L = self.lower_bound(pattern)
------- R = self.upper_bound(pattern)
------- return max(0, R - L)
```

9.0.8. *Booth Min String Rotation.*

```
def booth(s: str) -> int:
--- s = s + s
--- n = len(s)
--- f = [-1] * n
--- k = 0
--- for j in range(1, n):
------- i = f[j - k - 1]
------- while i != -1 and s[j] != s[k + i + 1]:
----------- if s[j] < s[k + i + 1]:
--------------- k = j - i - 1
----------- i = f[i]
------- if i == -1 and s[j] != s[k]:
----------- if s[j] < s[k]:
--------------- k = j
----------- f[j - k] = -1
------- else:
----------- f[j - k] = i + 1
--- return k
```

10. Other Algorithms

10.0.1. *Longest Increasing Subsequence.*

```
def lis(arr):
--- if not arr:
------- return []
--- n = len(arr)
--- seq = []
--- back = [0] * n
--- for i in range(n):
------- lo, hi = 0, len(seq)
------- while lo < hi:
----------- mid = (lo + hi) // 2
----------- if arr[seq[mid]] < arr[i]:
--------------- lo = mid + 1
----------- else:
--------------- hi = mid
------- res = lo
------- if res < len(seq):
----------- seq[res] = i
------- else:
----------- seq.append(i)
------- back[i] = -1 if res == 0 else seq[res - 1]
--- ans = []
--- at = seq[-1]
--- while at != -1:
------- ans.append(at)
------- at = back[at]
--- return ans[::-1]
```

10.0.2. *Cycle Find.*

```
def find_cycle(x0, f):
--- t = f(x0)
--- h = f(f(x0))
--- while t != h:
------- t = f(t)
------- h = f(f(h))
--- mu = 0
--- h = x0
--- while t != h:
------- t = f(t)
------- h = f(h)
------- mu += 1
--- lam = 1
--- h = f(t)
--- while t != h:
------- h = f(h)
------- lam += 1
--- return mu, lam
```

10.0.3. *2Sat.*

```
class TwoSAT:
--- def __init__(self, n):
------- self.n = n
------- self.N = 2 * n
------- self.adj = [[] for _ in range(self.N)]
--- def var(self, x):
------- return x << 1
--- def neg(self, x):
------- return x ^ 1
--- def add_imp(self, x, y):
------- self.adj[x].append(y)
--- def add_or(self, x, y):
------- self.add_imp(self.neg(x), y)
------- self.add_imp(self.neg(y), x)
--- def sat(self):
------- n = self.N
------- adj = self.adj
------- index = 0
------- stack = []
------- onstack = [False] * n
------- ids = [-1] * n
------- low = [0] * n
------- comp = [-1] * n
------- self.time = 0
------- self.comp_id = 0
------- def dfs(u):
----------- nonlocal index
----------- ids[u] = low[u] = index
----------- index += 1
----------- stack.append(u)
----------- onstack[u] = True
----------- for v in adj[u]:
--------------- if ids[v] == -1:
------------------- dfs(v)
------------------- low[u] = min(low[u], low[v])
--------------- elif onstack[v]:
------------------- low[u] = min(low[u], ids[v])
----------- if low[u] == ids[u]:
--------------- while True:
------------------- v = stack.pop()
------------------- onstack[v] = False
------------------- comp[v] = self.comp_id
------------------- if v == u:
----------------------- break
--------------- self.comp_id += 1
------- for i in range(n):
----------- if ids[i] == -1:
--------------- dfs(i)
------- for i in range(0, n, 2):
----------- if comp[i] == comp[i ^ 1]:
--------------- return False
------- return True
```

10.0.4. *Dpll.*

```
class SAT:
--- def __init__(self):
------- self.clauses = []
------- self.n = 0
--- def var(self):
------- self.n += 1
------- return self.n
--- def clause(self, c):
------- self.clauses.append(c)
--- def clause_sat(self, clause, assign):
------- unknown = False
------- for x in clause:
----------- v = abs(x)
----------- val = assign[v]
----------- if x > 0 and val == 1:
--------------- return True
----------- if x < 0 and val == 0:
--------------- return True
----------- if val == -1:
--------------- unknown = True
------- return False if not unknown else None
--- def propagate(self, assign):
------- changed = True
------- while changed:
----------- changed = False
----------- for clause in self.clauses:
--------------- sat = False
--------------- unassigned = []
--------------- for x in clause:
------------------- v = abs(x)
```

```
------------------- if x > 0 and assign[v] == 1: ---------------------
----------------------- sat = True ------------------------------------
----------------------- break -----------------------------------------
------------------- if x < 0 and assign[v] == 0: ---------------------
----------------------- sat = True ------------------------------------
----------------------- break -----------------------------------------
-------------------------------------------------------------------------
------------------- if assign[v] == -1: -------------------------------
----------------------- unassigned.append(x) --------------------------
-------------------------------------------------------------------------
--------------- if sat: -----------------------------------------------
------------------- continue ------------------------------------------
-------------------------------------------------------------------------
--------------- if len(unassigned) == 0: ------------------------------
------------------- return False  # contradiction ---------------------
-------------------------------------------------------------------------
--------------- if len(unassigned) == 1: ------------------------------
------------------- x = unassigned[0] ---------------------------------
------------------- v = abs(x) ----------------------------------------
------------------- val = 1 if x > 0 else 0 ---------------------------
-------------------------------------------------------------------------
------------------- if assign[v] != -1 and assign[v] != val: ----------
----------------------- return False ----------------------------------
-------------------------------------------------------------------------
------------------- if assign[v] == -1: -------------------------------
----------------------- assign[v] = val -------------------------------
----------------------- changed = True --------------------------------
-------------------------------------------------------------------------
------- return True -----------------------------------------------------
-------------------------------------------------------------------------
--- def solve_dfs(self, assign, idx=1): -------------------------------
------- if idx > self.n: ----------------------------------------------
----------- return True -------------------------------------------------
-------------------------------------------------------------------------
------- if assign[idx] != -1: -----------------------------------------
----------- return self.solve_dfs(assign, idx + 1) ----------------------
-------------------------------------------------------------------------
------- for val in [0, 1]: --------------------------------------------
----------- new_assign = assign[:] --------------------------------------
----------- new_assign[idx] = val ---------------------------------------
-------------------------------------------------------------------------
----------- if self.propagate(new_assign): ------------------------------
--------------- if self.solve_dfs(new_assign, idx + 1): -------------------
------------------- return True ---------------------------------------
-------------------------------------------------------------------------
------- return False ----------------------------------------------------
-------------------------------------------------------------------------
--- def solve(self): --------------------------------------------------
------- assign = [-1] * (self.n + 1) ----------------------------------
-------------------------------------------------------------------------
------- if not self.propagate(assign): --------------------------------
----------- return False ------------------------------------------------
-------------------------------------------------------------------------
------- return self.solve_dfs(assign) ---------------------------------
-------------------------------------------------------------------------
--- def get_value(self, x): -------------------------------------------
------- return self.assign[x] == 1 ------------------------------------
```

10.0.5. *Stable Marriage.*

```
from collections import deque -------------------------------------------
-------------------------------------------------------------------------
def stable_marriage(n, m_pref, w_pref): ---------------------------------
--- w_rank = [[0] * n for _ in range(n)] ------------------------------
-------------------------------------------------------------------------
--- for w in range(n): ------------------------------------------------
------- for i, man in enumerate(w_pref[w]): ---------------------------
----------- w_rank[w][man] = i ------------------------------------------
-------------------------------------------------------------------------
--- q = deque(range(n)) -----------------------------------------------
-------------------------------------------------------------------------
--- at = [0] * n ------------------------------------------------------
-------------------------------------------------------------------------
--- eng = [-1] * n ----------------------------------------------------
-------------------------------------------------------------------------
--- res = [-1] * n ----------------------------------------------------
-------------------------------------------------------------------------
--- while q: ----------------------------------------------------------
------- m = q.popleft() -----------------------------------------------
-------------------------------------------------------------------------
------- while at[m] < n: ----------------------------------------------
----------- w = m_pref[m][at[m]] ----------------------------------------
----------- at[m] += 1 --------------------------------------------------
-------------------------------------------------------------------------
----------- if eng[w] == -1: --------------------------------------------
--------------- eng[w] = m --------------------------------------------
--------------- res[m] = w --------------------------------------------
--------------- break -------------------------------------------------
-------------------------------------------------------------------------
----------- if w_rank[w][m] < w_rank[w][eng[w]]: ------------------------
--------------- q.append(eng[w]) --------------------------------------
--------------- eng[w] = m --------------------------------------------
--------------- res[m] = w --------------------------------------------
--------------- break -------------------------------------------------
-------------------------------------------------------------------------
--- return res --------------------------------------------------------
```

10.0.6. *Dates.*

```
def c_div(a, b): --------------------------------------------------------
--- return int(a / b) -------------------------------------------------
-------------------------------------------------------------------------
def date_to_int(y, m, d): -----------------------------------------------
--- k = c_div(m - 14, 12) ---------------------------------------------
-------------------------------------------------------------------------
--- return ( ----------------------------------------------------------
------- 1461 * (y + 4800 + k) // 4 ------------------------------------
------- + 367 * (m - 2 - k * 12) // 12 --------------------------------
------- - 3 * ((y + 4900 + k) // 100) // 4 ----------------------------
------- + d - 32075 ---------------------------------------------------
--- ) -----------------------------------------------------------------
-------------------------------------------------------------------------
-------------------------------------------------------------------------
def int_to_day(jd): -----------------------------------------------------
--- return jd % 7 -----------------------------------------------------
-------------------------------------------------------------------------
-------------------------------------------------------------------------
def int_to_date(jd): ----------------------------------------------------
--- x = jd + 68569 ----------------------------------------------------
--- n = 4 * x // 146097 -----------------------------------------------
--- x -= (146097 * n + 3) // 4 ----------------------------------------
-------------------------------------------------------------------------
--- i = (4000 * (x + 1)) // 1461001 -----------------------------------
--- x -= 1461 * i // 4 - 31 -------------------------------------------
-------------------------------------------------------------------------
--- j = 80 * x // 2447 ------------------------------------------------
--- d = x - 2447 * j // 80 --------------------------------------------
-------------------------------------------------------------------------
--- x = j // 11 -------------------------------------------------------
--- m = j + 2 - 12 * x ------------------------------------------------
--- y = 100 * (n - 49) + i + x ----------------------------------------
-------------------------------------------------------------------------
--- return y, m, d ----------------------------------------------------
```

10.0.7. *Simulated Annealing.*

```
import random -----------------------------------------------------------
import math -------------------------------------------------------------
import time -------------------------------------------------------------
-------------------------------------------------------------------------
-------------------------------------------------------------------------
def simulated_annealing(n, seconds): ------------------------------------
--- rng = random.Random() ---------------------------------------------
-------------------------------------------------------------------------
--- sol = list(range(1, n + 1)) ---------------------------------------
--- rng.shuffle(sol) --------------------------------------------------
-------------------------------------------------------------------------
--- def score(arr): ---------------------------------------------------
------- return sum(abs(arr[i] - arr[i - 1]) for i in range(1, n)) ----
-------------------------------------------------------------------------
--- cur = score(sol) --------------------------------------------------
-------------------------------------------------------------------------
--- T0 = 100.0 --------------------------------------------------------
--- T1 = 0.001 --------------------------------------------------------
--- start = time.time() -----------------------------------------------
--- iters = 0 ---------------------------------------------------------
-------------------------------------------------------------------------
--- while True: -------------------------------------------------------
------- if iters % 16 == 0: -------------------------------------------
----------- progress = (time.time() - start) / seconds ------------------
----------- if progress >= 1.0: -----------------------------------------
--------------- break -------------------------------------------------
----------- temp = T0 * (T1 / T0) ** progress ---------------------------
-------------------------------------------------------------------------
------- a = rng.randint(0, n - 2) -------------------------------------
-------------------------------------------------------------------------
------- x, y = sol[a], sol[a + 1] -------------------------------------
-------------------------------------------------------------------------
------- delta = 0 -----------------------------------------------------
-------------------------------------------------------------------------
------- if a > 0: -----------------------------------------------------
----------- delta += abs(sol[a - 1] - y) - abs(sol[a - 1] - x) ----------
-------------------------------------------------------------------------
------- if a + 2 < n: -------------------------------------------------
----------- delta += abs(x - sol[a + 2]) - abs(y - sol[a + 2]) ----------
-------------------------------------------------------------------------
------- accept = delta <= 0 or rng.random() < math.exp(-delta / temp)
-------------------------------------------------------------------------
------- if accept: ----------------------------------------------------
----------- sol[a], sol[a + 1] = sol[a + 1], sol[a] ---------------------
----------- cur += delta ------------------------------------------------
-------------------------------------------------------------------------
------- iters += 1 ----------------------------------------------------
-------------------------------------------------------------------------
--- return cur, sol ---------------------------------------------------
```

10.0.8. *Fast Input Reading.*

```
import sys --------------------------------------------------------------
-------------------------------------------------------------------------
data = sys.stdin.buffer.read().split() ----------------------------------
it = iter(data) ---------------------------------------------------------
-------------------------------------------------------------------------
def next_int(): ---------------------------------------------------------
--- return int(next(it)) ----------------------------------------------
```

10.0.9. *Bit Hacks.*

```
def snoob(x): -----------------------------------------------------------
--- y = x & -x --------------------------------------------------------
--- z = x + y ---------------------------------------------------------
--- return z | (((x ^ z) // y) >> 2) ----------------------------------
```